DMP INTRO

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FEA – April 27th 2022

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OUTLINE

- Intro on Research Data Management planning and DMP requirements
- DMPs how-to:
  - DMPonline.be, DMP templates and guidance
  - DMP sections
- Q&A
- Example projects

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RESEARCH DATA MANAGEMENT PLANNING AND DMP REQUIREMENTS

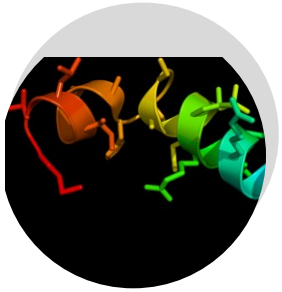
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RESEARCH DATA CAN MEAN A LOT

Survey



Measurement



Observation



Code



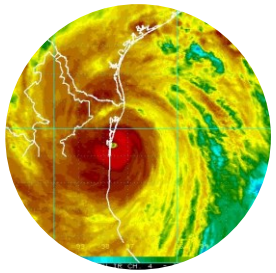
Image



Document



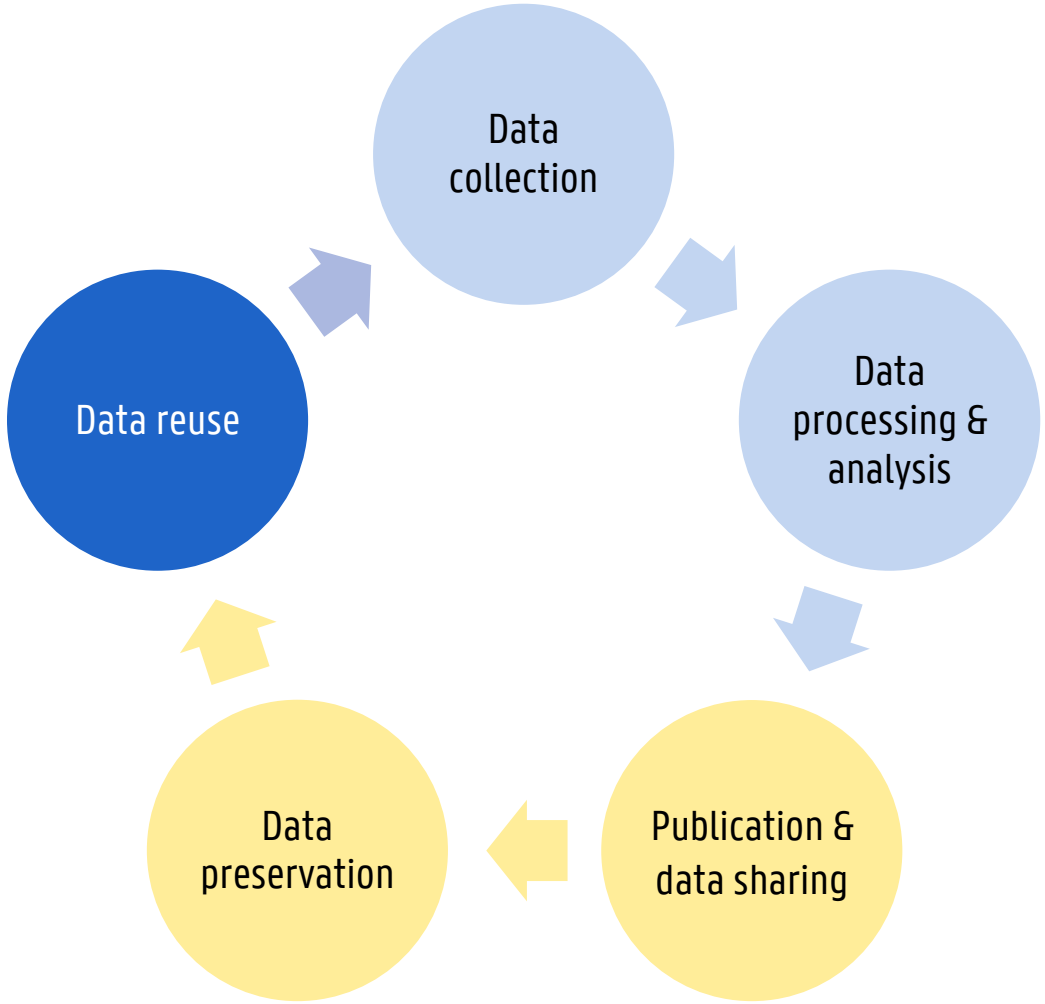
Visualisation



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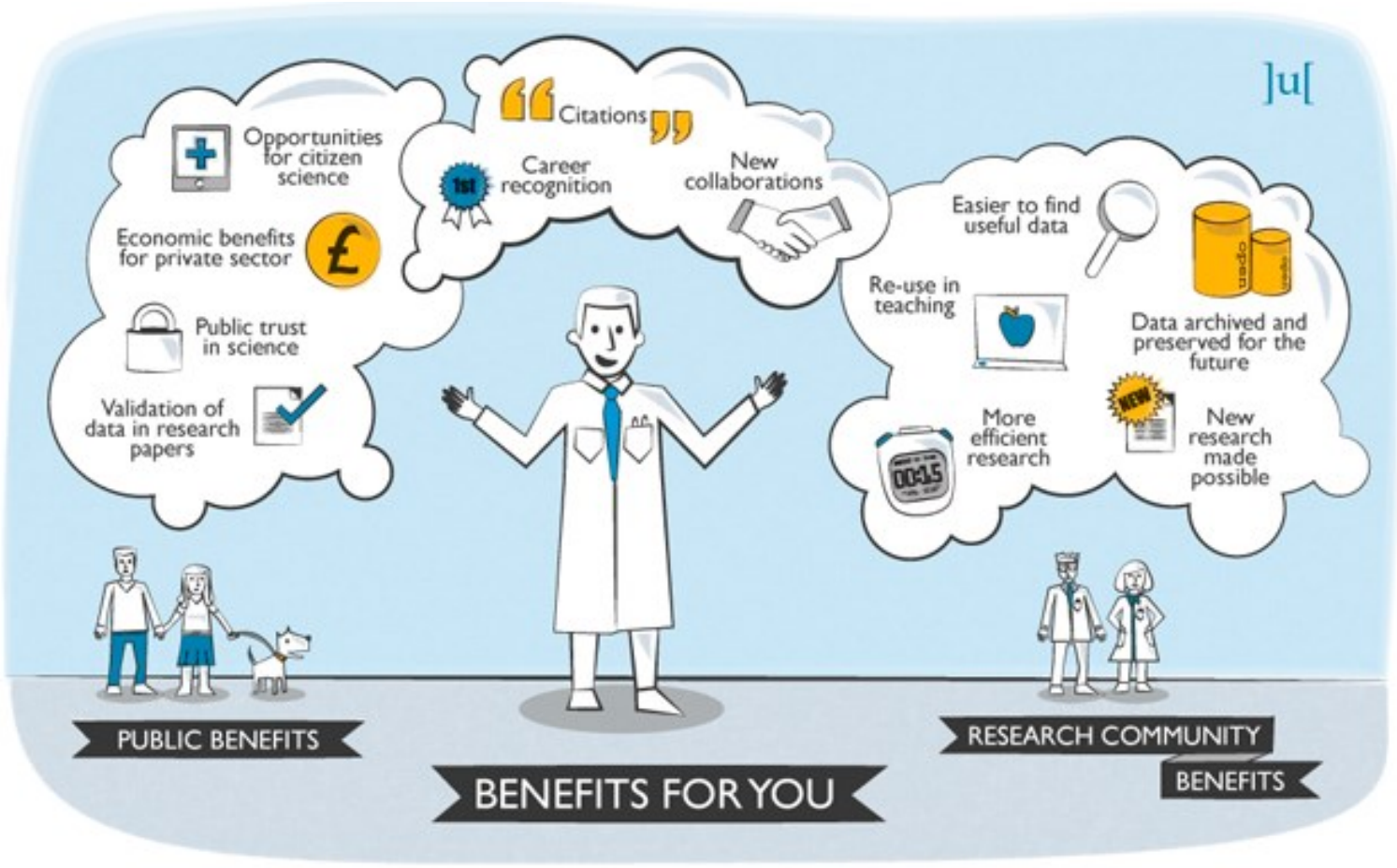
Research data is **any information** that has been collected, observed, generated or created **to validate original research findings.**

RESEARCH DATA LIFE CYCLE



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THE BENEFITS OF RDM, OPEN AND FAIR DATA



From 'Overcoming obstacles to sharing research data' by Brian Hole, licensed under CC-BY 4.0

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WITHOUT PLANNING RESEARCH DATA MANAGEMENT…

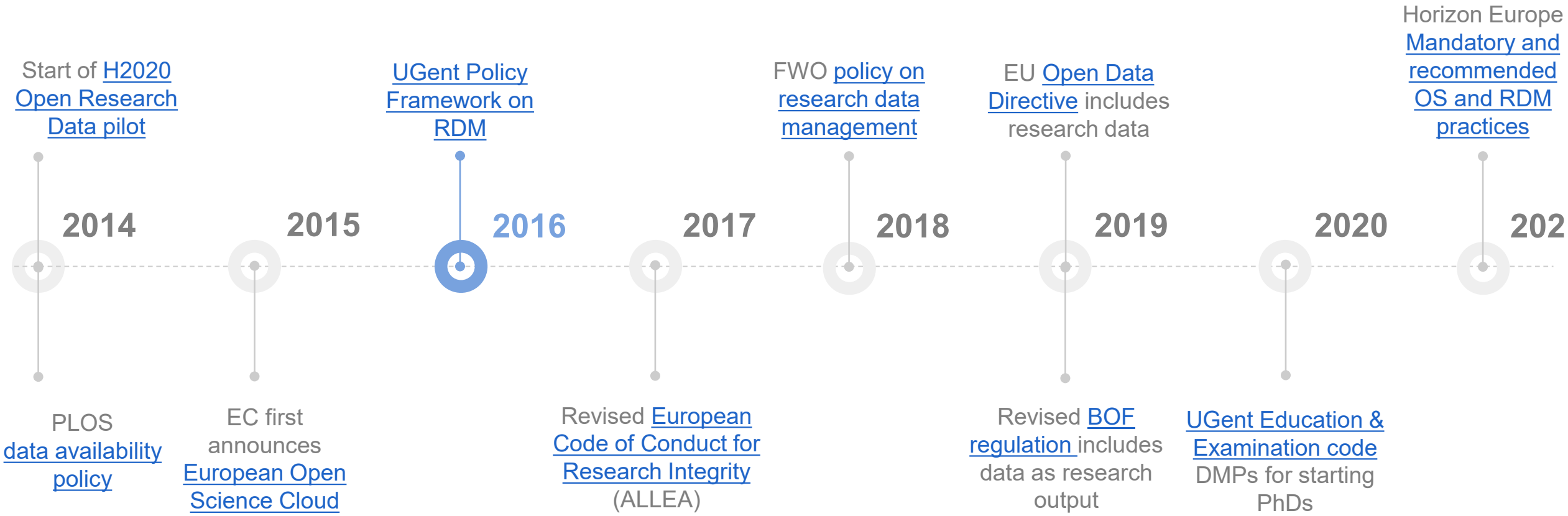
- teams/labs often lose data when researchers leave
- working with data from collaborators can be problematic
- there often is no budget allocated to maintaining data(bases) beyond a project's end
- gaining consent for archiving and sharing data is often forgotten
- …

→ DMPs take some time & effort upfront, but can save you headaches later on!

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CHANGING LEGAL & POLICY LANDSCAPE

Growing expectations of open & FAIR data



https://www.ugent.be/en/research/datamanagement/policies/overview.htm

EXTERNAL FUNDERS & UGENT: DMP AND RDM REQUIREMENTS

		HEurope	ERC	BELSPO	FWO	BOF / IOF (UGent)	PhD students (UGent)
DMP	Proposal	YES	NO	YES (Provisional DMP)	NO (5 RDM questions)	NO	NO
	Initial DMP	YES (By M6)	YES (By M6)	YES (By M6)	YES (By M6)	YES (By M6)	YES (By M6)
	Final DMP	YES	NO	YES	YES	YES	YES (prior to public defense)
	Periodic report	YES (Grant agreement)	NO	NO	NO	NO	NO
	Template*	YES (recommended)	YES (recommended)	YES (recommended)	YES (mandatory)	NO (use any template from DMPonline.be)	NO (use any template from DMPonline.be)
Other RDM	Open Data mandate	YES	YES	YES	NO	NO	NO
	Preservation period	NO	NO	NO	YES (≥ 5y)	YES (≥ 5y)	YES (≥ 5y)

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*All templates available via DMPonline.be

More info about external funders and UGent DMP requirements for new PhDs.

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1. Write a Data Management Plan
2. Adequately document data
3. Securely store & process data
4. Preserve relevant data for min. 5 years
5. Provide access to data within UGent + make them openly available (unless legitimate reasons for restricting data sharing)

https://www.ugent.be/en/research/datamanagement/policies/rdm-policy.pdf/at_download/file

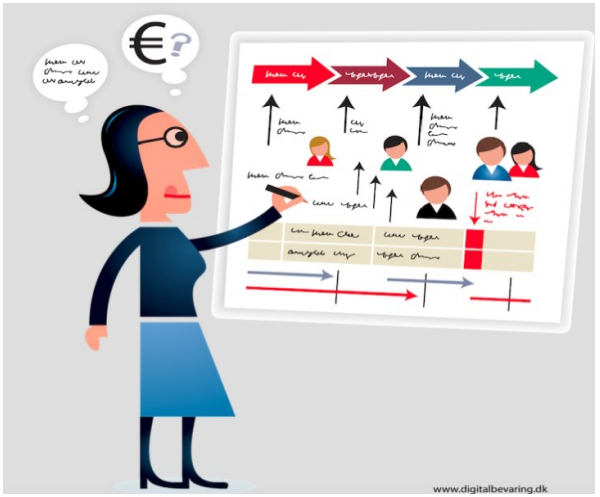
© UGent, foto Hilde Christiaens

DMP'S - GOOD PRACTICE FOR ANY PROJECT

Consider it a tool to help you

- increase awareness of RDM
- ensure data are of high quality & remain usable
- comply with legal, ethical & other requirements
- identify any support & resources required
- clarify expectations & responsibilities
- starting point for a data paper?

But of course: a plan cannot do the work for you!

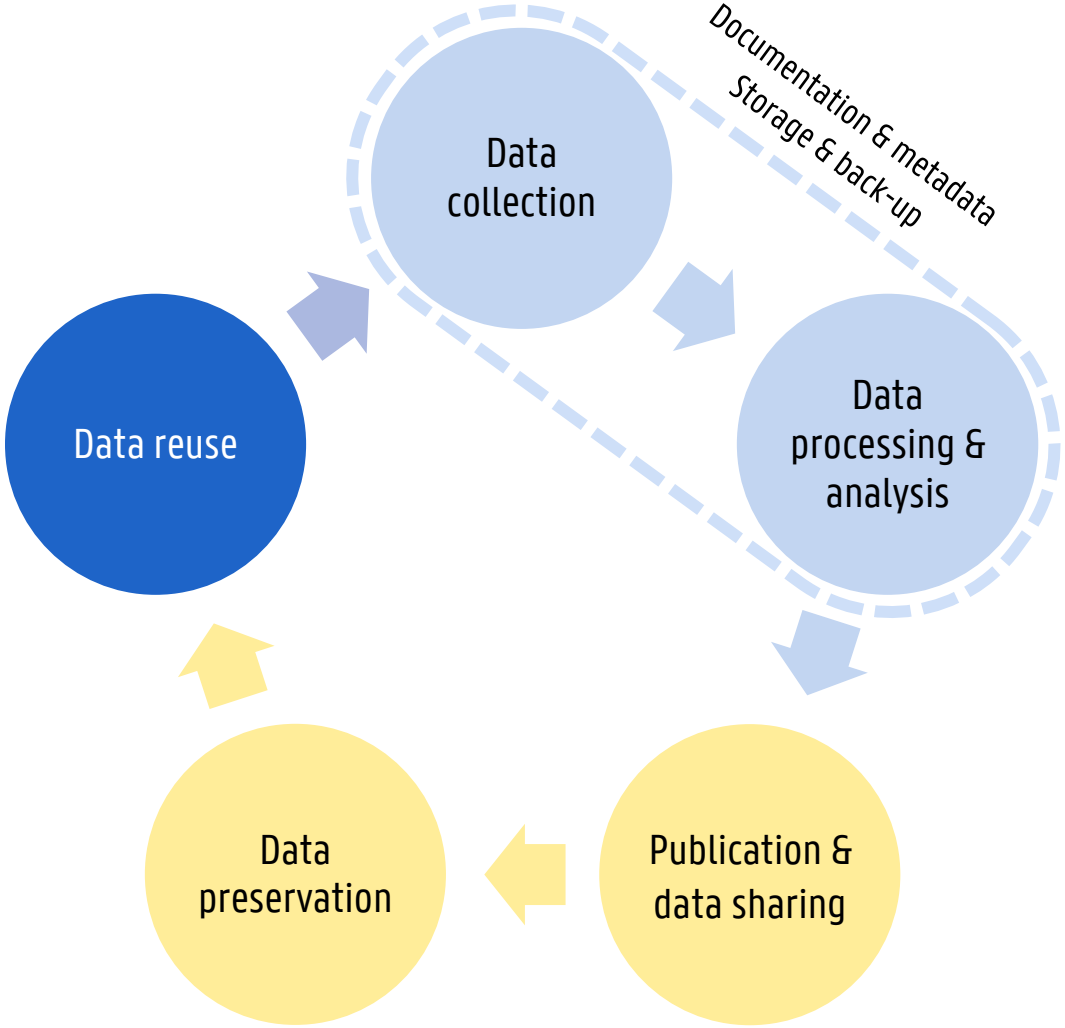


'Planning' by Jørgen Stamp, attribution: digitalbevaring.dk, licensed under CC-BY 2.5 DK

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DATA MANAGEMENT PLAN (DMP)

- Outlines how research data will be handled **during** & **after** a project.



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HOW TO WRITE A DMP?

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DMPONLINE.BE TUTORIAL

https://youtu.be/5U9_DDdBzYk



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Click 'Create plan(s)'

My Dashboard

The table below lists all the plans associated with the **current user account**. This includes the plans you have created with this account and the plans that have been shared with you for this account.

Project Title	Template	Edited	Role	Test	Visibility	Shared	
Interesting collaborative project	Generic DMP +	14-09-2021	Owner	☐	Private	Yes	Actions
My European project	Horizon 2020 FAIR DMP +	14-09-2021	Owner	☐	Private	No	Actions
Demo project	FWO DMP +	14-09-2021	Owner	☐	Private	No	Actions
<fsqdvqfvqfdv	GDPR Record	16-08-2021	Owner	☑	N/A	No	Actions

Create plan

DMP ONLINE .BE | My Dashboard | Create plans | Reference | Help | Stefanie De Bodt

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www.ugent.be | Send mail to request DMP feedback (attach pdf!)

Ghent University RDM webpages | rdm.support@ugent.be

Admin

Create a new plan

Before you get started, we need some information about your research project to set you up with the best DMP template for your needs.

* What research project are you planning?

☐ mock project for testing, practice, or educational purposes

* Select the primary research organisation

Organisation

Ghent University (UGent)

- **or** - ☐ No research organisation associated with this plan or my research organisation is not listed

* Select the primary funding organisation

Funder

fwo

Fonds voor Wetenschappelijk Onderzoek - Research Foundation Flanders (FWO)

- **or** - ☐ No funder associated with this plan or my funder is not listed

Create plan | Cancel

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Create a new plan

Before you get started, we need some information about your research project to set you up with the best DMP template for your needs.

* What research project are you planning?

☐ mock project for testing, practice, or educational purposes

* Select the primary research organisation

Organisation

Ghent University (UGent)

- **or** - ☐ No research organisation associated with this plan or my research organisation is not listed

* Select the primary funding organisation

Funder

Begin typing to see a list of suggestions.

- **or** - ☑ No funder associated with this plan or my funder is not listed

Which DMP template would you like to use?

AVG Record (NL)

We found multiple DMP templates corresponding to your funder.

AVG Record (NL)
Faculty of Law & Criminology DMP +
Faculty of Psychology & Educational Sciences DMP +
GDPR Record (EN)
Generic DMP +

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Demo project

Project Details | Plan overview | Application DMP | Full DMP + | GDPR Record | DPIA | Share | Download

expand all | collapse all

21/29 answered

1. General Information (3 / 3) +

2. Data description (2 / 2) +

3. Ethical and legal issues (4 / 4) +

4. Documentation and metadata (1 / 2) −

What documentation will be provided to enable understanding and reuse of the data collected/generated in this project?

B *I*

The study will be documented by recording information on experimental setup, treatments, observations and measurements as well as analysis and calcuation methods.

- Methods will be recorded in an electronic notebook as plain text.
- Experimental setup will be recorded in a csv table containing details on the plants (Plant number, Community_type, Site, Block, Plot, N_Treatment, Herbivore_treatment, Species Status, Species Family)

Datasets will be documented by providing sufficient information on provenance of the data as part of the raw data tables to complement the methods in plain text.

Variables will be documented by generating a codebook listing the variables (columns names in the raw data table), their definition, storage type, possible values, an missing value code).

Save

Guidance | Comments

UGent

Documentation −

Good documentation provides the descriptive and contextual information necessary to properly find, interpret and use data in the future.

Identify early on what types of documentation you will need to provide and the efforts involved, taking into account your preservation and sharing plans for the data. You will typically need to document

- at the research project/study level: e.g. research design, context of data collection, data

MAIN DMP SECTIONS

Data summary / description

Purpose, origin, type, format & volume of the data

Ethical and legal issues

IRB approval, IP, GDPR

Documentation and metadata

Where and how, standards, ontologies, vocabularies

Resources

Costs



Responsibilities

Owner and roles

Data storage during the project

Security, back-up, transfer & access

Data sharing and reuse

Repositories, terms, contracts, licenses

Long-term preservation

What, where, restrictions?

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DATA DESCRIPTION QUESTIONS

- Describe the origin, type and format of the data (per dataset) and its estimate volume.
- Will you generate/collect new data and/or make use of existing data?
- What (if any) costs are involved in reuse of data, and to what extent is reuse and sharing of these data allowed (e.g. by checking licenses)?

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WHICH CATEGORIES OF DATA?

CONTENT OR OUTPUT TYPE

- Text / qualitative
- Numeric / quantitative
- Multimedia
- Code

METHOD

- Observational
- Experimental
- Simulation
- Derived or compiled…

FORMAT

- Open vs. closed

NATURE

- Digital (digital born or digitized)
- Analogue

ORIGIN

- Primary
- Secondary

STAGE

- Raw
- Processed

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DATA OVERVIEW: DIFFERENT APPROACHES

Write a data description paragraph

- Straightforward project structure
- Limited number of data types

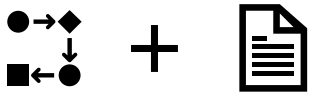
Build a project data workflow

- Complex project structure, small to medium number of data types.
- To visualize data as input & outputs + the different processing and analysis steps.
- Can help in other RDM steps: file organization & naming, documentation.

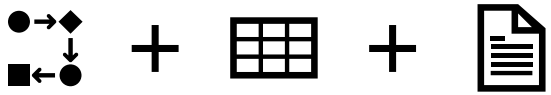
Prepare a data summary table

- Complex project structure, medium to large number of data types.
- Data sets and/or types as rows, with characteristics as columns.
- To detect RDM needs: e.g. additional storage space needed, confidential data…









DATA OVERVIEW

- Straightforward project structure, limited number of datatypes

⇨ Write a description

EXAMPLE

Types of data

Qualitative (e.g. phenotyping data for cell lines), quantitative (e.g. cell counts), mass spectrometry and image data will be generated. Raw data will be analysed and expressed as graphs, tables and annotated images, some of which, it is expected, will be published.

Format and scale of the data

Data generated will be in various formats and sizes of datasets, all of which will be accessible using common software allowing easy access and long term validity during and after the project, thus facilitating data sharing. The format/types of data include:
i) Cell images e.g. phase and fluorescence, and electron micrographs (~5,000 images over project). Software used includes OpenLab, Softworx and IN Cell Investigator, with data saved as software-specific files e.g liff and lg3 files, as well as generic formats such as jpeg, tiff etc.

….

https://www.gla.ac.uk/media/Media_442573_smxx.pdf

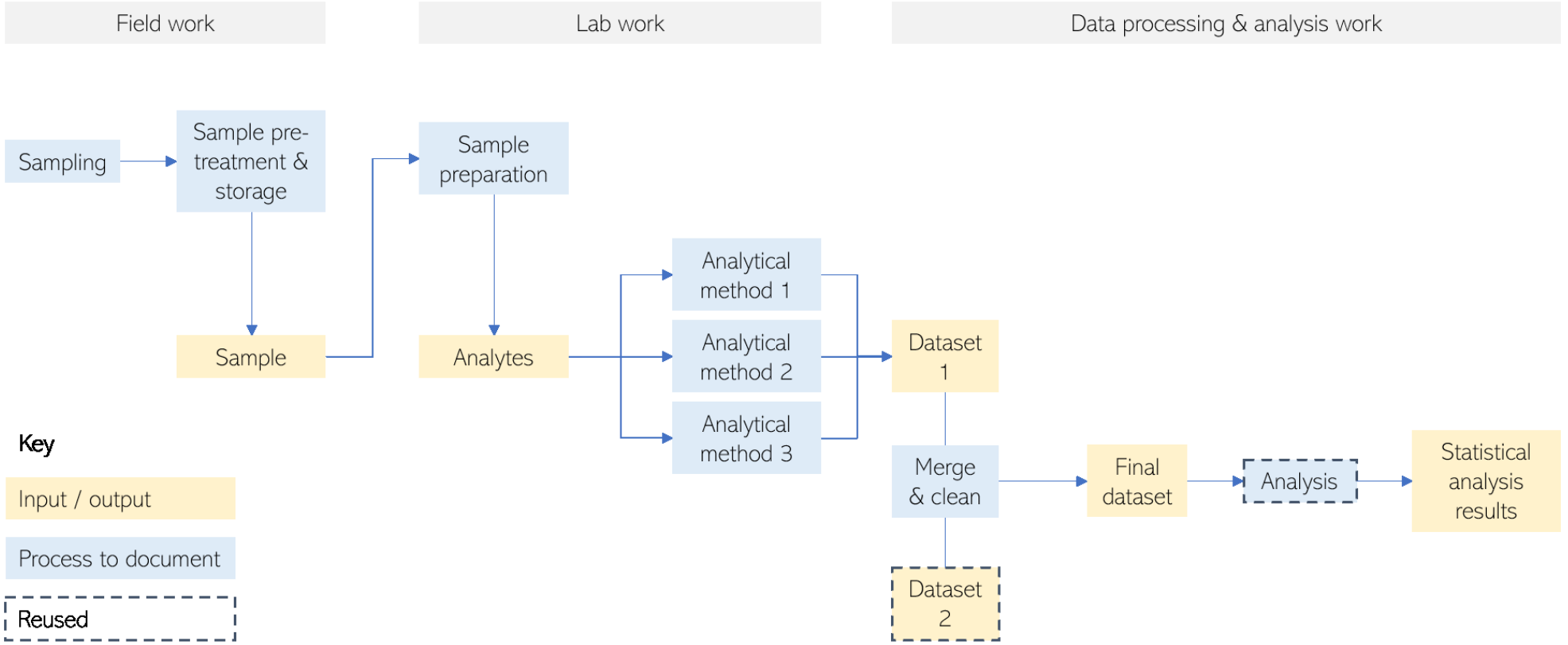
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DATA OVERVIEW

- Complex project structure, small to medium number of datatypes

⇒ Design a data flow, add details in text or table

EXAMPLE



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DATA OVERVIEW

- Complex project structure, medium to large number of datatypes

⇨ Create a data table

EXAMPLE

Study phase	Data type	Origin	Format	Expected Volume
WP1	Observational qualitative	Questionaire for older adults using momentary assessment	Paper and pencil + digitized transcript (.docx)	<1 GB, paper surveys of 300 participants
WP2	Experimental	Micro-randomized trial to determine the effect of isolated determinants on physical activity	Digital textual and numerical data combined in spreadsheets and obtained via a smartphone-based experiment. CSV format and GT3X for raw activity tracker data	<5 MB for the CSV files. Raw accelerometer data (.GT3X files) are 250kB per participant x 40 participants per experimental study = 10MB
WP3	…	…	…	…

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ETHICAL & LEGAL ISSUES QUESTIONS



Are there any ethical issues concerning the creation and/or use of the data (e.g. experiments on humans or animals, or dual use)? Add the reference to approval by ethical committee(s).



Is there potential for tech transfer and valorization? Will IP restrictions be claimed for the data you created? If so, for what data, and which restrictions will be asserted?



Do existing 3rd party agreements restrict dissemination or exploitation of the data you (re)use? If so, to what data do they relate and what restrictions are in place?



Will you use personal data? If so, briefly describe the kind of personal data you will use.



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ETHICAL & LEGAL ISSUES – SOME TIPS (I)

Ethical issues

→ answer 'yes' if e.g.

- you need permission from ethical committee and/or via informed consent for collecting & handling data (incl. archiving & sharing data)
- you work with confidential or otherwise sensitive data that need to be handled confidentially (i.e. data that could cause harm if disclosed – e.g. to endangered species, vulnerable archeological sites, national security, sensitive personal data…)

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ETHICAL & LEGAL ISSUES – SOME TIPS (II)

IP & third-party agreements

→ restrictions on collecting, sharing, archiving & reusing data may be applicable, especially if

- your research has commercial potential: to valorise results, sharing may not be possible (to protect confidential knowhow) or only after embargo period (to seek patent protection first) – always contact UGent TechTransfer before disclosing anything!
- you work with third-party data/collaborate with external partners (e.g industry): check licenses/agreements to know what you can do

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ETHICAL & LEGAL ISSUES – SOME TIPS (III)

Personal data

→ = any information relating to an identified or identifiable (directly or indirectly) living individual

DMP: Will you use personal data?

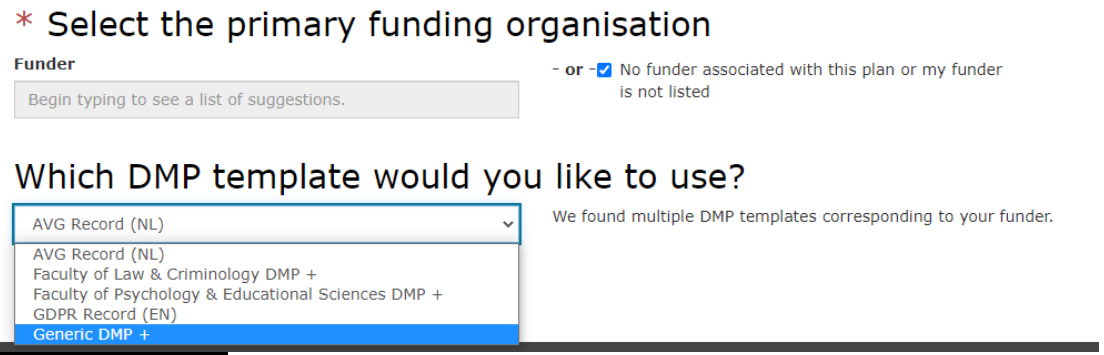
- If so, briefly describe the kind of personal data you will use.
  - Should be coherent with (& can refer to) data overview section
- Will you need informed consent or permission for collecting, using, preserving and/or sharing your data?
- How will you protect research participants' identities if required (e.g. anonymization, controlled access)?
- How will you ensure secure transfer and storage of confidential data?
  - Should be coherent with (or refer to) data storage & security questions

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ETHICAL & LEGAL ISSUES – SOME TIPS (IV)

DMP ≠ Register of processing activities

- GDPR (General Data Protection Regulation) requires that all personal data processing activities are documented and registered in a **'register of processing activities'**, (aka GDPR Register).
- GDPR register can be done with DMPonline.be, in a GDPR record tab, but it is not part of the DMP.
- GDPR register has to be done before personal data is processed, regardless whether a DMP is required or not.
- See https://researchtips.ugent.be/en/tips/00001795/



GDPR record only



DMP + GDPR record only

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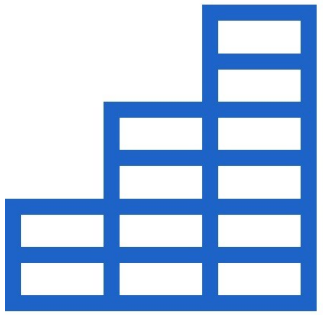
DATA STORAGE & SECURITY QUESTIONS



- Where will the data be stored?
- Is there sufficient storage capacity?
- What are the expected costs for data storage during the project? How will these costs be covered?
- How will the data be backed up? What provisions are in place for data recovery?
- How will you control access to the data (ensure data cannot be accessed or modified by unauthorized persons?
- How will you ensure secure transfer of data?

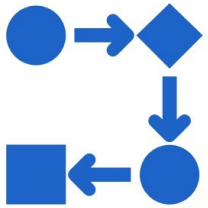
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FACTORS INFLUENCING DATA STORAGE SOLUTION



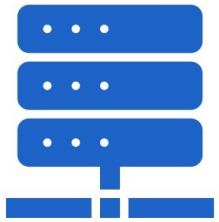
VOLUME

which total data volume do you expect to generate or collect?



ACTIVE

do you want to store data that you are actively working on or do you want to preserve data?



COMPUTING PERFORMANCE

do you need high performance computing for the analysis of the data?



TIME

how long does the data need to be stored and remain available?



ACCESS

who needs access to the data and how (online/offline)?



CONFIDENTIALITY

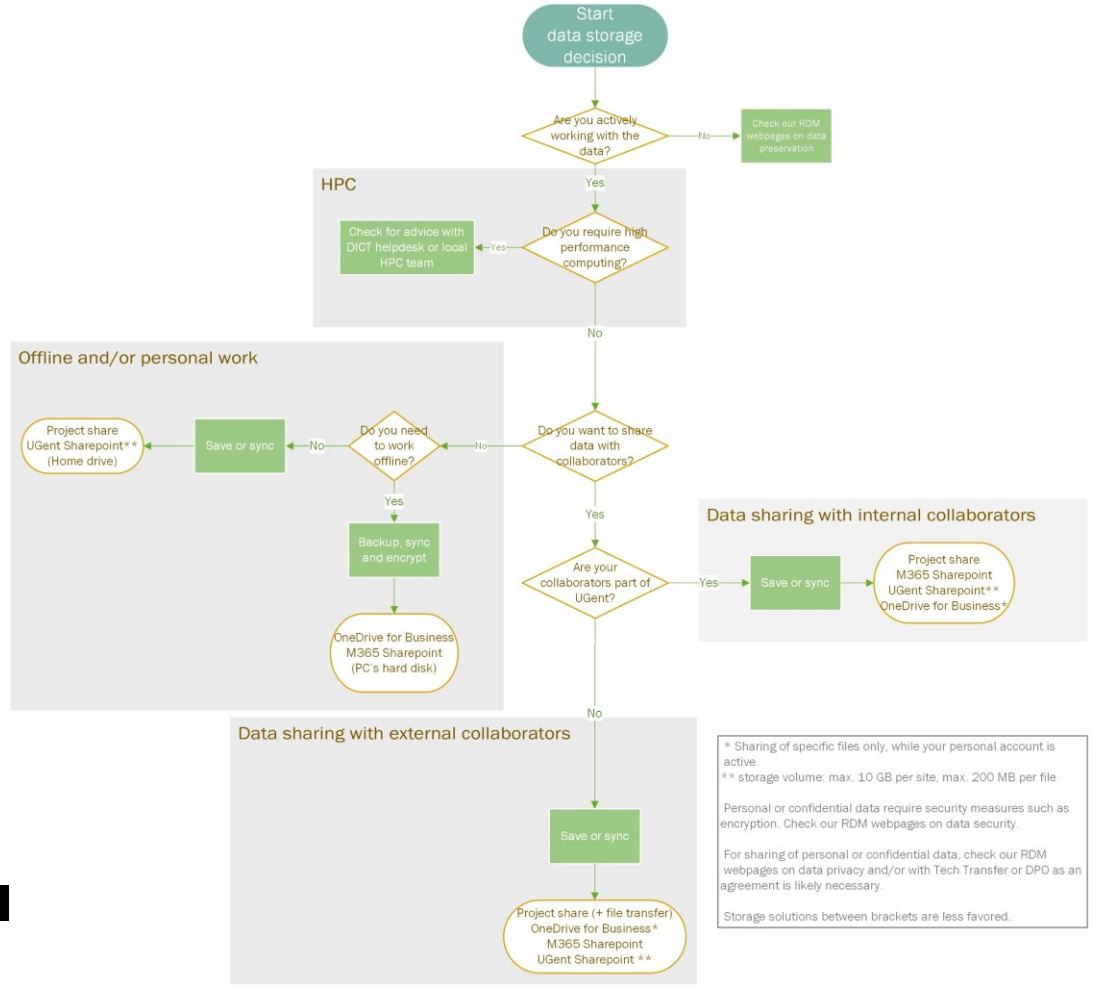
are security measures needed to safeguard the data?

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	Microsoft 365		Sharepoint	Shares		HPC (only showing temp storage)	
Name	OneDrive FB	Teams SharePoint	UGent Sharepoint	H:/	Project Share (acl & bulk)	HOME/DATA/SCRATCH	VO_DATA/VO_SCRATCH
Location	Cloud	Cloud	UGent	UGent	UGent	UGent	UGent
Size	3TB per user	25 TB per site/Team	10 GB per site	5 GB	GB-TB range	25 GB	GB-TB range
Backup mechanism	Versioning	Versioning	Versioning	snapshots	snapshots	none	none
Sync to local	✓	✓					
Confidential	Encrypt	Encrypt	✓	✓	✓	✓	✓
Intended use	Personal	Sharing	Sharing	Personal	Sharing	Personal	Sharing
Sharing external users?	✓	✓	✓				✓
Suitable for TB datasets					✓		✓
Data in use	✓	✓	✓	✓	✓	✓	✓
Data at rest					✓		

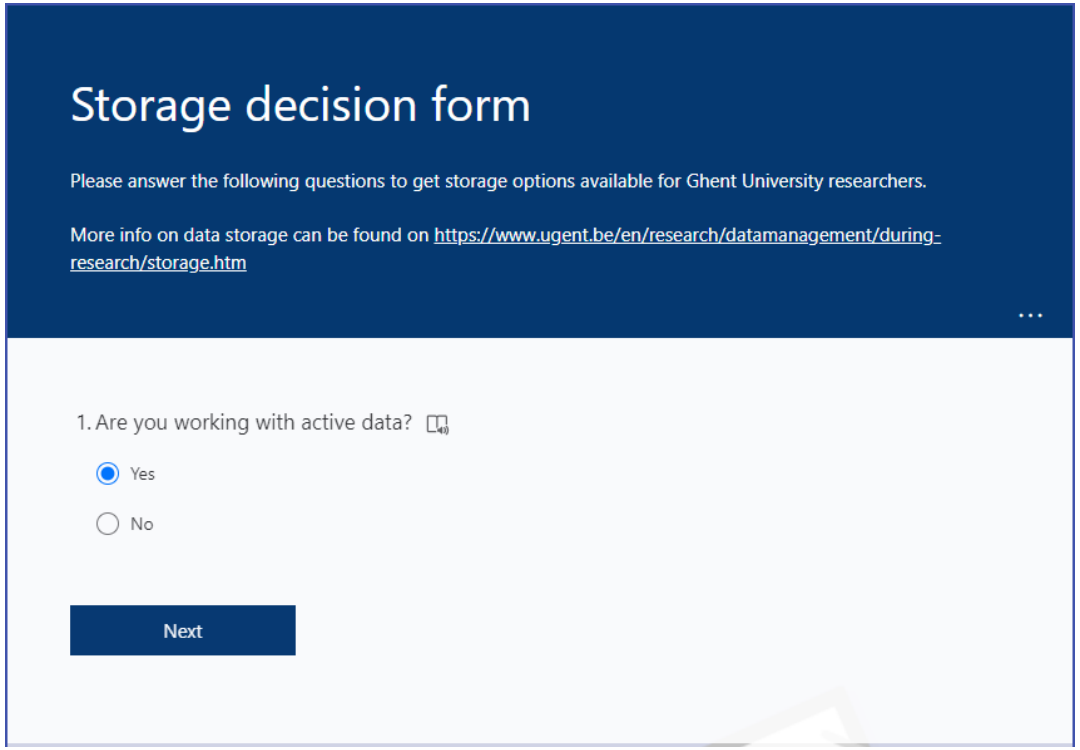
CHOOSE APPROPRIATE STORAGE SOLUTION

Decision Tree



https://osf.io/z73ng/

Decision Form



https://forms.office.com/r/AH3UBC5j3X

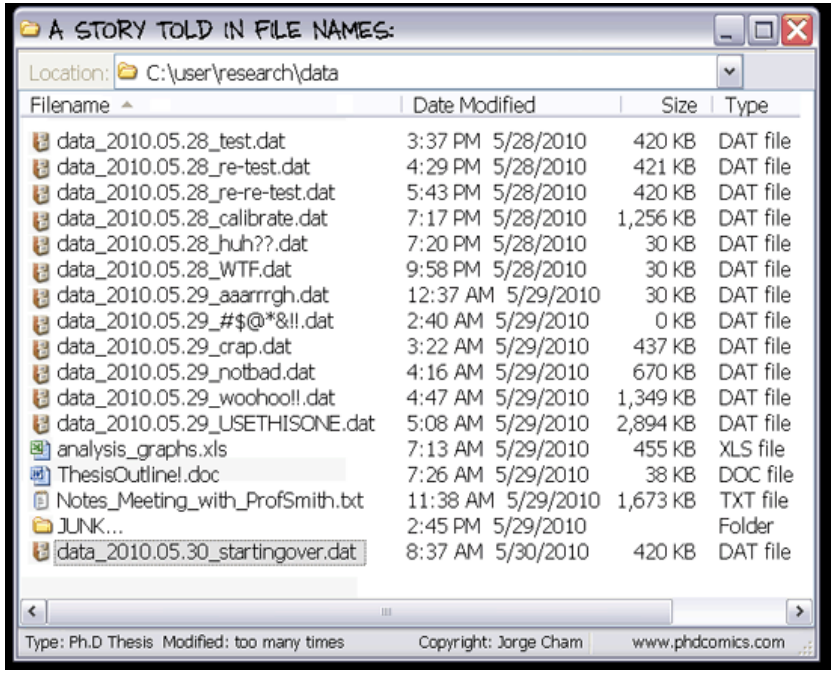
FILE ORGANIZATION: WHAT AND WHY

What?

- File naming convention
- Folder structure
- Version control
- Document your decisions!

Why?

- Efficiency: to locate a file more easily
- Collaboration
- Avoid data loss
- Keep files understandable and reusable



Harvard RDM working group, CC-BY (https://osf.io/hjntb/)

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Knowledge clip: Keeping research data organized

CONSIDER THE DO'S AND DON'TS OF DATA SECURITY

Avoid local storage or own IT resources

Avoid storing confidential information in third-party cloud services

Do not send data by email

Use institutional network storage

Backup regularly

Configure access permissions

Encrypt confidential data

Use safe file transfer alternatives

Keep your system updated

Use strong passwords

Protect your hardware

Control physical access to confidential data

When needed, destroy information in a safe way

- Shred paper
- Erasing software
- Destroy hardware

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More: https://www.ugent.be/en/research/datamanagement/during-research/data-security.htm

Knowledge clip: Data security

DATA DOCUMENTATION & METADATA QUESTIONS

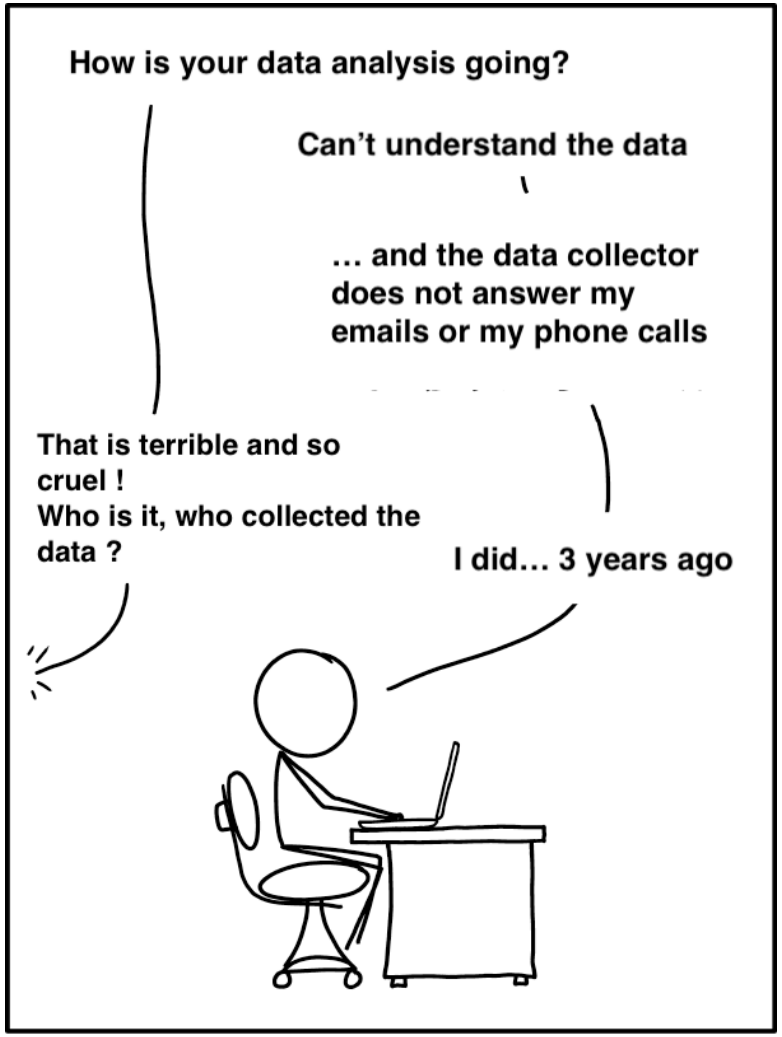


- Review & verification of research
- Reproducibility
- Make data understandable and reusable

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WHY DOCUMENTING DATA?

- Help science:
  - Research review & verification
  - The reproducibility crisis
- Help yourself & others:
  - “Your future self will be your first collaborator”
  - Will you remember or will others understand…
    - what your variables mean?
      - E.g. "sglmemgp" = single member of group
    - the procedures used to transform or derive variables?
  - Keep data understandable & reusable



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LEVELS OF DATA DOCUMENTATION

DOCUMENTATION LEVEL	WHAT TO DOCUMENT	WHERE TO DOCUMENT
Item Variable	Methods, units, possible values, synonyms, abbreviations	Codebook, data dictionary
Sample Study object	Study object, method, instrument settings, time, place	(Register) table, protocol, notebook
File or database	File inventory and relations, database structure	Readme, database schema, data listings (qualitative data)
Project or study	Subject, grant, contributors, methods, process, workflow	Notebook (e-, code), document, protocol, readme, data inventory table

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WHAT WHEN WHERE HOW WHO WHY

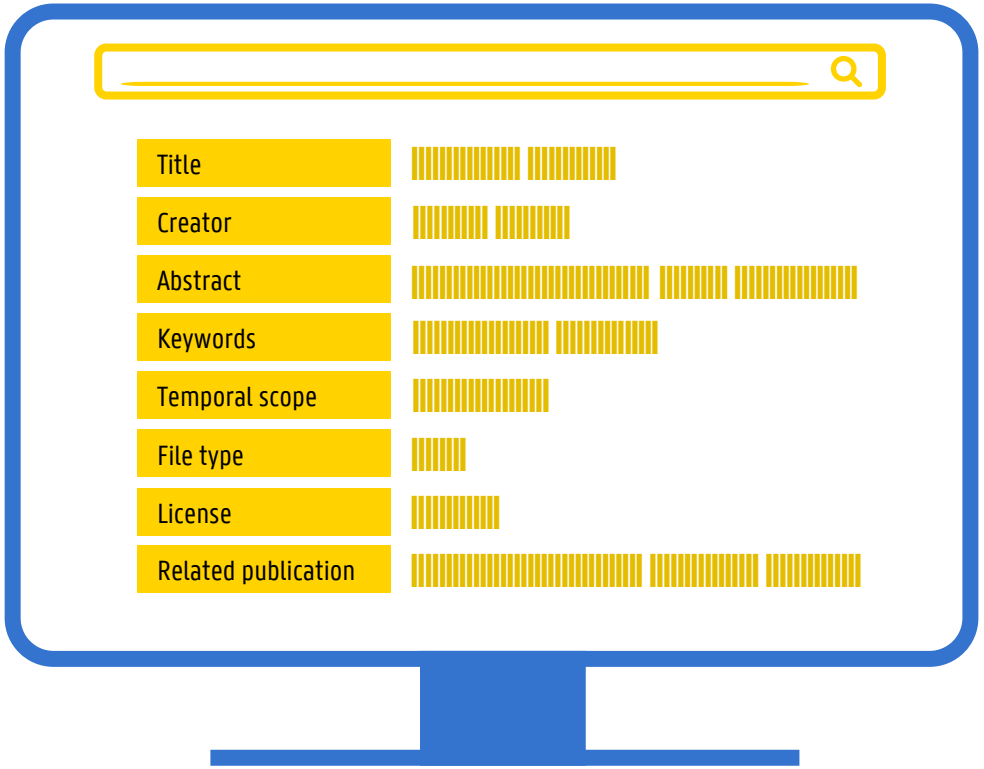
DOCUMENTATION

Any contextual information that allows data to be understandable and reusable.



METADATA

Information about data, recorded in a highly structured way, using a set of defined information fields or elements to be machine readable.



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EXAMPLE METADATA & STANDARDS

- The following standards should be used for data documentation:
  - IEC TS 62600-100 [7] : technical specification on power performance assessment of electricity producing wave energy converters.
  - IEC TS 62600-200 [8] : Electricity producing tidal energy converters - Power performance assessment.
  - ISO 14224:2006 [9] : collection and exchange of reliability and maintenance data for equipment
  - …
- Metadata records will accompany the data files in order to describe, contextualise and facilitate external users to understand and reuse the data.

- DTOceanPlus will adopt the DataCite Metadata Schema [10] , a domain agnostic metadata schema, as the basis for harvesting and importing metadata about datasets from data archives.

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DTOCEAN+: https://phaidra.univie.ac.at/open/o:1139995

EXAMPLE METADATA & STANDARDS

The following metadata should be created to identify datasets:

- **Identifier**: A unique string that identifies the dataset.
- **Author/Creator**: The main researchers involved in producing the data in priority order.
- **Title**: A name or title by which a data is known.
- **Publisher**: The name of the entity that holds, archives, publishes prints, distributes, releases, issues, or produces the data.
- **Publication Year**: The year when the data was or will be made publicly available.
- **Subject**: Subject, keyword, classification code, or key phrase describing the resource.
- **Access rights**: Provide a rights management statement for the dataset. Include embargo information if applicable.
- **Geo-location**: Spatial region or named place where the data was gathered
- ….

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DTOCEAN+: https://phaidra.univie.ac.at/open/o:1139995

DATA PRESERVATION QUESTIONS

Data preservation

≠

Not deleting data files

- Which data will be retained for the expected 5-year period after the end of the project?
- Where will these data be archived (= stored for the long term)? List options if not clear yet
- What are the expected costs for data preservation during these 5 years? How will the costs be covered?

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WHAT TO PRESERVE? SELECTION CRITERIA

Not all data might need to be kept, especially after the mandatory retention period.

1. What is required?
    a) Legal or ethical requirements: what and for how long.
    b) Institutional, funder, journal policies. E.g. UGent & FWO = min. 5 years.
2. For which purposes are you preserving data?
    a) Research verification
    b) Future reuse
    c) Other (e.g. education)
3. Uniqueness and value of data
4. Cost-benefit

You know your data best!

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More: https://www.dcc.ac.uk/guidance/how-guides/five-steps-decide-what-data-keep

WHERE TO PRESERVE DATA?

- Research data and documentation selected for retention should be kept in a **suitable location** and in a secure manner to ensure that they remain available and usable beyond the end of your project, with appropriate access rights, and in sustainable formats.
- Suitable location -> main purpose of preservation? (verification, internal or external reuse).

Data repository

- Recommended / preferred practice.
- *The* option when goal is sharing & reuse.

In house

- Sometimes it is not be possible/appropriate to deposit data in an external repository, e.g. for legal, ethical, contractual, practical, or other reasons.
- Data needs to be preserved for verification purposes
- Best in-house option: project shares.

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DATA SHARING & REUSE QUESTIONS

- Which data will be made available after the end of the project?
- Where/how and when will the data be made available for reuse?

- Motivate why data needs restricted or closed access
- Who will be able to access the data and under what conditions?

- What are the expected costs for data sharing?

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WHERE TO SHARE DATA?

Recommended: a discipline-specific data repository

Instead of share upon email request, share on project website, share as supplementary material



DATA FINDABILITY FAIR



RESEARCH COMMUNITY SUPPORT



PERSISTENT & UNIQUE IDENTIFIER FAIR



LONG TERM RESERVATION FAIR



DATA ACCESS CONDITIONS FAIR



DATA RE-USE CONDITIONS FAIR

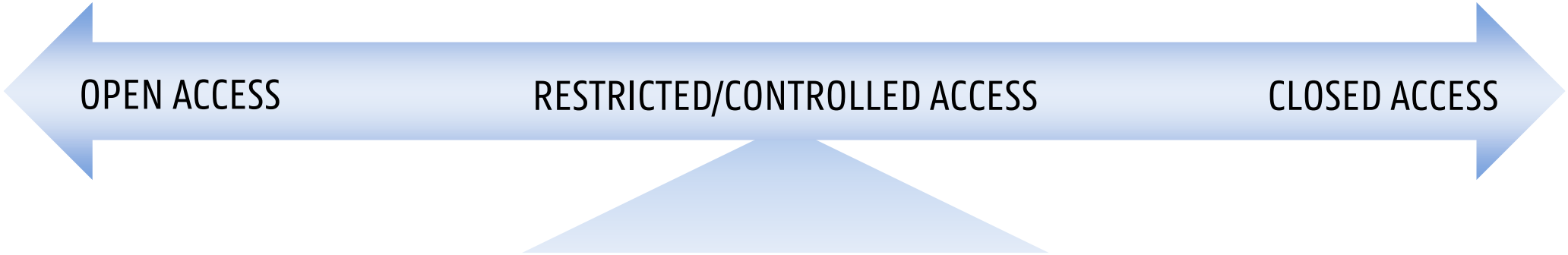
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FAIR: Findable, Accessible, Interoperable, Re-usable

HOW TO SHARE DATA? – ACCESS LEVELS

What about sharing of sensitive or confidential data? (cfr ethical or legal issues)

"As open as possible, as closed as necessary"



Can be freely used, modified & shared by anyone for any purpose

- Personal data
- Confidential data
- Sensitive data
- 3rd party data
- Copyright/database right
- Valorization potential

Under embargo
Unable to share

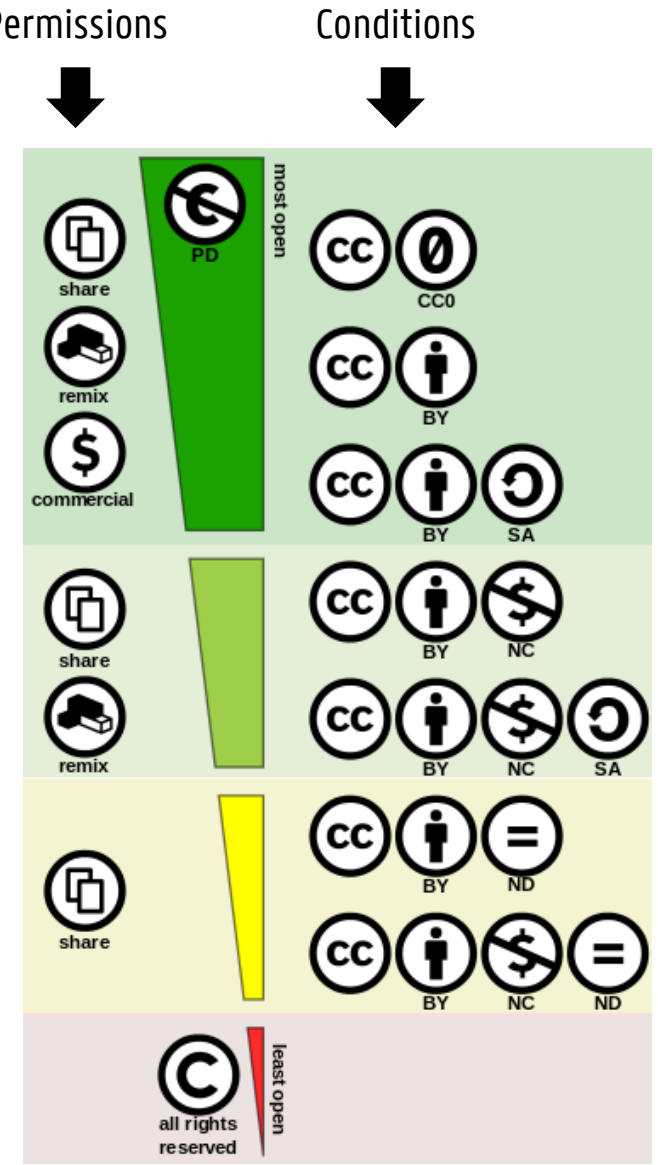
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WHICH RE-USE IS ALLOWED? - LICENSES

Creative Commons (CC) Licenses

Grant permissions under certain conditions:

- **Attribution (BY)**: give proper credit to the original creator
  - → Stacking, long chain of derivations, manual inspection of contributions.
- **Copyleft (Share Alike)**: license the new (derived) creations under the identical terms (same license).
  - → Can be difficult to fulfil.
- **Non-commerciality**: to prevent the new (derived) work to be commercially exploited.
  - → Subject to interpretation ("commercial use"): e.g. journal can make profit while author doesn't.
- **Non-derivatives**: no adaptations can be made.
  - → Strong limitation for dataset reuse and/or integration..



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MAIN DMP SECTIONS

Data summary / description

Purpose, origin, type, format & volume of the data

Ethical and legal issues

IRB approval, IP, GDPR

Documentation and metadata

Where and how, standards, ontologies, vocabularies

Resources

Costs



Responsibilities

Owner and roles

Data storage during the project

Security, back-up, transfer & access

Data sharing and reuse

Repositories, terms, contracts, licenses

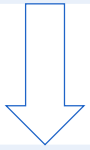
Long-term preservation

What, where, restrictions?

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DMP: FIRST VERSION VS. LAST VERSION

Data Management Plan
First submission



Data Management Plan
Final submission

- Basic information about your data.
- Data storage and back up vs. long term preservation.
- Initial idea about metadata standards and dedicated archive / repository.
- Identification of legal and/or ethical issues.
- Establishing responsibilities.
- First scope of reusability.

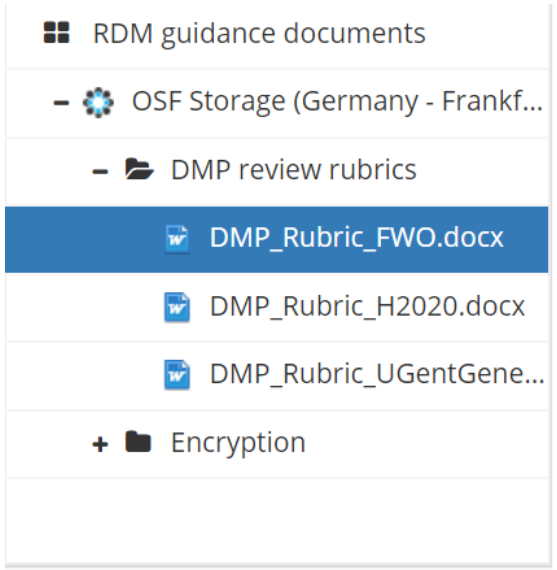
- Precise description of all data types.
- Clear long-term preservation strategy: selected data archive(s) and/or repository(ies).
- FAIR data and open data ("As open as possible, as closed as necessary").
- Definite selection of reusable outputs.

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Adapted from: Jasmin Böhmer. (2019, September). UBC RDM Workshop 1 - Data Management Planning (Version V2). Zenodo. http://doi.org/10.5281/zenodo.3469953

SELF-ASSESSMENT GRIDS AND SUPPORT

- Self assessment grids (FWO template and others): https://osf.io/dhpr3/
- Contact your data steward at rdm.support@ugent.be



Version 2019-11-25

FWO DMP Evaluation Rubric

This is an evaluation rubric for the DMP template of the FWO (i.e. the FWO template for DMPs that have to be drafted for funded projects, not the RDM questionnaire that is part of the FWO application form). It is inspired by and based on the example rubric (CC0) from the DART project (https://osf.io/26b9r/), which was presented during the IDCC 2016 DMP Workshop in Amsterdam on 25 February 2016.

	PERFORMANCE LEVELS		
PERFORMANCE CRITERIA	**Complete/detailed**	**Addressed issue, but incomplete/too vague**	**Did not address issue**
1. Indicates whether the project will generate/collect **new data** and/or make use of **existing data**. **[Q2.1]**	Clearly indicates whether the project will generate new data and/or reuse existing data, and explains why. E.g.: *"The project will reuse existing datasets on local mobility in the city of Ghent. These data will provide sufficient information to perform a quantitative analysis of the use of public transport services. However, to allow a qualitative analysis of local perceptions of public transport, the project will also generate new data, as there are no existing qualitative data about that aspect of our study."* E.g.: *"The geological data generated in the 'Ancient*	Indicates whether the project will generate new data and/or reuse existing data, but does not or only vaguely explain why.	There is no indication as to whether newly generated and/or existing data will be used.

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DATA STEWARDS - SERVICES: ADVISORY & TRAINING

DMP support

templates, guidance, examples, feedback



Online information & guidance

via RDM webpages, (re)search tips, knowledge clips…



RDM helpdesk

rdm.support@ugent.be

for all RDM-related questions



Consultations



Training, info sessions & other events

User support for Library RDM tools

e.g. DMPonline.be, OSF for UGent

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Q&A

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EXAMPLES

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PROJECT 1 – FWO TEMPLATE

- WP1: Investigate thermochemical treatment techniques for nutrient recycling and pollutant removal from sewage sludge
- WP2: Investigate consumer attitudes regarding food grown using fertilizers based on nutrient recycled from sewage sludge

- Inspired on publications https://doi.org/10.1080/10643389.2018.1558889, https://doi.org/10.1016/j.wasman.2008.09.011 and dataset https://data.mendeley.com/datasets/kccc8m9pn9/1

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Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

Data summary / description

Purpose, origin, type, format & volume of the data

Will you generate/collect new data and/or make use of existing data?

- [x] Generate new data
- [ ] Reuse existing data

Comment

- Only new data will be generated during the project.
- WP1: The setup of the different thermochemical treatments will be based on results from previous research which will be appropriately credited at time of publication.
- WP2: data will be generated by means of a questionnaire which will be prepared by the project researchers.

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

Describe the origin, type and format of the data (per dataset) and its (estimated) volume.

- WP1.
  - Sewage sludge ash will undergo thermochemical treatment under a systematic variation of parameters: composition and concentration of chemical reactors, temperature and retention time. For each of the treatments, we will measure:
    - Pollutant removal: comparison of the concentration of heavy metals and organic pollutants before and after the treatment.
    - Improvement of nutrient bio-availability: we will compare nutrient bio-availability under different treatments to determine the influence of the dependent variables (chemical treatment, temperature, retention time).
  - Data will be compiled in excel spreadsheets using templates prepared in advance in order to facilitate data entry and avoid mistakes. Data will be migrated to csv for analysis, which will be performed in R.
  - The purpose of these data is to find the most efficient thermochemical treatment(s), where pollutants are removed below the permitted legal limits, while producing high nutrient availability rates.
- WP2. Data on attitudes towards consumption of food that has been grown using fertilizers based on recycled human excreta will be investigated by means of an online questionnaire.
  - Online questionnaires will be performed using Lime Survey. Results will be exported as csv. The analysis will be performed in SPSS (*.sav).
  - The purpose of these data is to evaluate the willingness or unwillingness of potential consumers, and to find potential explanatory factors. The results can be used to design e.g. awareness raising campaigns or stakeholder engagement actions, in preparation of public consultation procedures for the implementation of nutrient recycling methods.

WP	Data type	Format	Estimated size
WP1	Pollutant concentration before and after treatment	xlsx, csv	<5MB
WP1	Nutrient bio-availability	xlsx, csv	<5MB
WP1	Data from statistical analysis	R	<5MB
WP2	Consumers attitudes - questionnaire responses	csv	<5MB
WP2	Consumer attitudes - analysis	sav	<5MB

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Ethical and legal issues

IRB approval, IP, GDPR

Will you use personal data? If so, shortly describe the kind of personal data you will use AND add the reference to your file in your host institution's privacy register.

- (●) Yes
- ( ) No

Additional Information

Short description of the kind of personal data that will be used:

- We will collect demographic data from participants: place of residence, age, gender, education, occupation, annual income.
- Besides, the questionnaire will collect the respondent's opinions or views about issues in relation to the research topic: environmental awareness questions, attitudes or acceptance of the proposed product, etc.

The legal ground for processing data is consent.

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

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Ethical and legal issues

IRB approval, IP, GDPR

Does your work possibly result in research data with potential for tech transfer and valorisation? Will IP restrictions be claimed for the data you created? If so, for what data, and which restrictions will be asserted?

- (●) Yes
- ( ) No

Comment

The thermochemical treatment(s) developed in WP1 of the project could have commercialization potential. We will investigate how to seek IP protection of our findings with the UGent Tech Transfer team. All data from WP1 will be therefore treated as confidential until the appropriate IP protection measures have been taken (e.g. patent application).

Do existing 3rd party agreements restrict dissemination or exploitation of the data you (re)use? If so, to what data do they relate and what restrictions are in place?

- ( ) Yes
- (●) No

Comment

[redacted]

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

Data storage during the project

Security, back-up, transfer & access

Where will the data be stored?

- Data will be stored in a dedicated share that will be requested for the project.
- Data will be organized according to WP.

How will the data be backed up?

Project shares are maintained by UGent ICT department, regular backups are foreseen. Snapshots of previous versions or deleted files or folders can be retrieved.

Is there currently sufficient storage & backup capacity during the project? If yes, specify concisely. If no or insufficient storage or backup capacities are available, then explain how this will be taken care of.

(•) Yes
( ) No

Additional Information

The default storage volumes of the project shares are sufficient for the volume that is expected to be generated by this project. Storage costs are covered by the University's central ICT department.

What are the expected costs for data storage and backup during the project? How will these costs be covered?

No additional costs are foreseen.

Data security: how will you ensure that the data are securely stored and not accessed or modified by unauthorized persons?

- Access to the share and permissions on folders can be managed. Only researchers working on the project, as well as the supervisor will have access (and write permissions) to the project share.
- Only the researcher and the supervisor working on WP2 have access to the specific folder containing survey results, as well as to the Lime Survey record. Lime Survey is a tool supported by UGent.

Note: for an additional layer of security, data could be encrypted. Encryption is recommended when dealing with confidential data, and strictly necessary when storing such data in cloud services.

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

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Documentation and metadata

Where and how, standards, ontologies, vocabularies

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

What documentation will be provided to enable understanding and reuse of the data collected/generated in this project?

- WP1.
  - The details of the **experimental design** will be thoroughly documented in a word document. This includes the instrumentation used at the different phases of the thermochemical treatment (e.g. oven or condensation tube types), as well as the subsequent laboratory analysis to quantify concentration of pollutants and nutrients. These will be made available as supplementary material in a research paper, after having secured IP protection.
  - For each experiment, treatment operational parameters (chemical reactors, temperature and retention time) will be documented on a stand-alone spreadsheet. We will also document the date of the experiment, researcher(s) involved.
  - Before performing the experiments, a **data entry template** will be prepared in excel. This will contain a table with the necessary columns to fill in the experiment

Will a metadata standard be used? If so, describe in detail which standard will be used. If not, state in detail which metadata will be created to make the data easy/easier to find and reuse.

(●) Yes
( ) No

Additional Information

- We will keep (at minimum) the following metadata: Title, abstract, authors and affiliation, keywords and project information. We will prepare a **readme** template that will be kept together with the data files of each WP, based on existing examples: https://cornell.box.com/v/ReadmeTemplate
- At the time of publishing, we will explore appropriate repositories and provide metadata in accordance to their guidelines and, therefore, **adhering to the standard the repository uses** (e.g. Dublin Core). The actual repository is not yet known. The **DMP will be updated once a decision has been made**.

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Long-term preservation

What, where, restrictions?

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

Which data will be retained for the expected 5-year period after the end of the project? In case only a selection of the data can/will be preserved, clearly state the reasons for this (legal or contractual restrictions, physical preservation issues...).

All data and documentation from both WPs will be retained for the minimum 5-year preservation period.

Where will these data be archived (= stored for the long term)?

The data in project shares will be moved to an archive folder. The structure and access permissions will remain the same.

At the end of the project, prior to publication and archiving, files under proprietary formats (e.g SPSS) will be **migrated to open formats**, to ensure long-term sustainability of the files.

Data from WP1 will also be available through a **data repository (potentially after an embargo period**). Data from WP2 will be made available through a restricted access repository. (See next section for more details).

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Data sharing and reuse

Repositories, terms, contracts, licenses

Research objective: investigate techniques and consumer attitudes about nutrient recycling from sewage sludge.

Are there any factors restricting or preventing the sharing of (some of) the data (e.g. as defined in an agreement with a 3rd party, legal restrictions)?

- ○ No
- ◉ Yes. Specify:

Additional Information

- WP1: the results from this WP might have **valorization potential**. We will consult with **UGent Tech Transfer** and explore the possibilities to secure our findings until they have been protected (e.g. by applying for a patent).
- WP2: **consent will be obtained** for all the personal data processing activities in the project, including data preservation and sharing.
  - Consent will be given by participants (and recorded) at the moment of participating in the online survey (alongside with information about the purpose of the project and the data collection). Participants will be informed that individual responses will not be used in a way that would allow their identification.
    - The survey is designed so that no data on direct identifiers are collected. Before publishing survey responses, we will evaluate if additional **anonymization** is needed (e.g. generalization).
  - To participate in the survey, respondents will be asked for consent for data publication. They will be informed about the possible reuses of the data that would be allowed, and where the data will be made available.

Where/how will the data be made available for reuse?

- ☑ In an Open Access repository
- ☑ In a restricted access repository
- ☐ Upon request by mail
- ☐ Other (specify):

Additional Information

- Data from WP1 can be made available in an OA repository after the appropriate IP protection has been guaranteed. This will be explored with UGent TechTransfer.
- Data from WP2 will be made available in a restricted access repository, to ensure that data can only be accessed under the terms specified by the consent.

[redacted]

When will the data be made available?

- ☐ Immediately after the end of the project
- ☑ Upon publication of the research results
- ☑ After an embargo period. Specify the length of the embargo and why it is necessary

Additional Information

- Data from WP1 might need an **embargo** period. We will explore how to protect our research results with the Tech Transfer team and update the DMP once a decision is made.
- Data from WP2 can be published at the **same time as research results**.

Who will be able to access the data and under what conditions?

- (WP1) Data to which no sharing restrictions apply will be shared openly, by assigning a **CC-BY v4.0 license** (and only after the embargo period if this is needed).
- (WP2) The access will be limited and will only be granted upon request, to the cases where the purpose of the reuse falls within the expectations of the data subjects and is compatible with consent from data subjects.
  - A **user agreement** will be necessary. We will explore drafting a bespoke license or agreement with Tech Transfer, or use data repositories specialized in the sharing of personal data.

PROJECT 2 – UGENT GENERIC TEMPLATE

- Develop a method to estimate surface soil moisture from satellite observations using neural networks
- No new data generated, method developed using existing data
- Research output: method performance, scripts, method

- Inspired on publication https://doi.org/10.1016/j.jhydrol.2019.124351

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Research objective: develop method to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

Data summary / description

Purpose, origin, type, format & volume of the data

What data will you collect or create?

The project will generate new data and reuse publicly available data.

- **Reused data:**
  - Ground-based SSM measurements from monitoring stations.
    - These data will be used to perform a spatial & temporal match on satellite data and to train the model.
    - Data are available in csv format, under CC-BY license, and total volume is in the order of 100 MB.
  - Satellite-based data:
    - Temperature brightness moisture, surface soil temperature, vegetation water content from satellite-based composites from SPL3SMP, GEOS-5 and MODIS.
    - These data are used to train the model and to produce the final SSM maps for the study area.
    - Data are available under open formats (e.g. HDF-5, HDF-EOS and NetCDF). Total volume of data will be in the range of 500 GB.
    - These data are reusable under a CC-BY license.
- **Own outputs:**
  - GNRR model:
    - Will be generated in Python (*.py). These are text files which are not expected to need more than 5 MB.
  - Model performance data.
    - The data will be in csv format, smaller than 100 KB ad will be used to select the best model.
  - SSM maps:
    - This will be the r
      entire study area
    - The maps will be
      below 20 GB in t

Dataset & type	Origin	Content/format	Estimated volume
Soil surface moisture (SSM) - Observational data	Open data collected in a set of ground-based monitoring stations that are run by different US public authorities and that are accessible at http://nationalsoilmoisture.com/. License: CC-BY	Format: csv (open format)	<100 KB per file. # necessary point data files = 500-1000. Total vol. = 50 - 100 MB
Soil moisture (SM) - Derived data	Publicly available satellite-based data from the SPL3SMP composite developed by NASA. Available at: https://nsidc.org/data/SPL3SMP. License: CC-BY	Format: HDF-5 (open format)	Approx. 30 MB per grid. # necessary grids in the order of 500-1000. Total volume: 15-30 GB
Surface soil temperature (Ts) - Derived data	(...) from GEOS-5 by NOAA. Available at: (...)	Format: HDF-5 and NetCDF-4 (open)	(...)
Vegetation water content (VWC) - Derived data	(...) from MODIS. Available at: (...)	Format: HDF-EOS (open format)	(...)
GNRR training - Development code	Python scripts to train the generalized regression neural network (GNRR) model, using previous datasets as input.	Format: python (.py)	<5 MB
Model evaluation scores - Simulation data	Quantitative (statistical) assessment of models, using 10-fold cross-validation technique.	Format: csv (open format)	<100 KB

Research objective: develop method to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

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Data summary / description

Purpose, origin, type, format & volume of the data

How will the data be collected or created?

- The data used to train the model (ground-based and satellite-based data) will be downloaded and stored in the project share, where a specific folder called "PhD_2020_2024_GNRR_SSM" has been created for the project.
- Within this folder, the different data and outputs are organized according to the table below.
- Data will be stored in the UGent central disk space (aka "project shares"). Data on the project shares is regularly backed up by the UGent ICT department, and retrieving snapshots is possible. Besides, most of the model input data is publicly available and can be downloaded again if necessary.
- The GNRR training model will be developed in Python and managed in GitHub (UGent GitHub), a dedicated tool for software development and version control.

Dataset	Folder	File naming system
Soil surface moisture (SSM) - Ground station	SSM_ground_stations	The file naming system used by data provider will be kept. File names indicate the parameter measured and date of measurement. E.g. "perc_05" = Daily soil moisture percentile (5cm depth). E.g. perc_05_20180801.csv
Temperature brightness (Tb) - Satellite	SPL3SMP_SM	The file naming system used by data provider will be kept (described at the user guide). File name provides information about, f.e. the mission data (e.g. SMAP), the specific composite product (e.g. L3_SM_P = Level 3, soil moisture, passive); date (YYYYMMDD), composite release id... E.g. SMAP_L3_SM_P_20210117_R17000_001.h5

Guidance | Comments (3)

UGent

Questions to consider:

- Are there any data capture standards or methodologies common in your discipline that you will use?
- What structure and naming system will you use for your files and folders?
- Do you have a strategy for version control, i.e. to track changes in files and retrieve a correct file version later?
- How will you ensure and check quality during and after data collection or creation?

Research objective: develop method to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

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Ethical and legal issues

IRB approval, IP, GDPR

How will you manage any ethics and confidentiality issues?

- There are no specific ethical or confidentiality issues.

How will you manage intellectual property rights issues?

- We do not own the rights of the data used to train the model. However, these data are publicly available in the specified sources and under the specified licenses provided in the Data Collection section of the DMP. These data will be appropriately referenced in the different research outputs or publications resulting from the project.
- The code will remain private / non-disclosed until the time of publishing. After that, it will be published under an open license to be yet decided (e.g. CC-BY-SA or GNU General Public License).

Research objective: develop method to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

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Data storage during the project

Security, back-up, transfer & access

Research objective: develop method to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

Data Storage and Backup during Research (2 / 2)

How will you store and backup data during research?

- All data will be stored in a dedicated project share from UGent central disk space.
- This infrastructure is maintained by UGent ICT department. There is sufficient storage capacity (capacity of the project shares is on demand), with no additional costs for the project. Content of the project shares is regularly backed up, with snapshots that can be retrieved up to 15 weeks.
- Certain calculations (e.g. model training) will be performed locally due to volume. Data will be synced daily with the projects shares using e.g. FreeFileSync.
- Code will be stored and managed with GitHub.

Guidance

UGent

Questions to consider:

- Where will you store the data during research?
- Do you have sufficient storage, or will you have to take additional measures?
- What is your backup strategy?
- How will the data be recovered after a loss?

How will you ensure that stored data are secure?

- Although no confidential data will be generated, only authorized people will have access to the project share (e.g. promotor).
- The personal computer that will be used to perform calculations will be password protected, and system will be regularly updated.
- The code will remain private / non-disclosed (in a private GitHub repository) until the time of publishing. Only the researcher and the promotor will have access to this repository.
- *Note: if data was confidential, it should be encrypted in your local disk.*

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Documentation and metadata

Where and how, standards, ontologies, vocabularies

How will you document the data?

Documentation:

- The **GNRR_SSM GitHub repository** will contain:
  - A **readme file** which will provide the following information: project name and description, instructions to install project locally and how to use it; the license. Once the project becomes public, instructions to contribute may be added as well.
  - The description will include contextual bibliographic references, e.g. to the underlying GNRR model and the model evaluation technique.
  - A **list of the reference datasets** that are used to train the model. For each data type, a link to the relevant documentation offered by the data provider will be given (e.g. user guidelines of SPL3SMP composite).
  - A **codebook** providing information about the relevant variables in the (ground-based and satellite) data used to train the model (variable name, description, units, possible values, etc). This information is also available via the data provider's documentation, but will be summarized in a table.
  - A codebook providing information about model evaluation score variables.
  - To ensure the code remains understandable and reusable, it will be thoroughly annotated and styled following the PEP 8 Style Guide for Python Code.

Metadata:

- At the end of the project, we intend to make the results openly accessible. We will publish our **repository** and link it to Zenodo to obtain a DOI. Rich metadata will be provided via Zenodo at the time of publishing (e.g. title, author(s), abstract, keywords, license, related publications, etc.). Metadata in Zenodo is compatible and can be exported into different standards (DataCite, DublinCore).
- By publishing our resuls in Zenodo, we make sure this will be findable, accessible, interoperable and reusable.
- We aim to publish the resulting SSM maps in Pangaea. The minimum and recommended metadata fields in Pangaea are available at: https://wiki.pangaea.de/wiki/Metadata

Research objective: develop model to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

Based on: https://doi.org/10.1016/j.jhydrol.2019.124351

Long-term preservation

What, where, restrictions?

Which data should be retained for preservation and/or sharing?

- All data in the project will be retained for **at least 5 years** (preservation period required by UGent policy).
  - The data used to train the GNRR model is already publicly available and will be therefore not be kept longer than this period. References to the specific datasets will still be kept, but the files will be deleted.
- The **code will be available via GitHub** (and Zenodo) for at least 5 years after the project.
- The results of the model evaluation will also be kept in the project shares, although a copy of this file will also be provided via Zenodo.
  - Intermediate / temporary files that might be created during model development will not be kept.

What is the long-term preservation plan for the selected datasets?

- A copy of the datasets used to develop the model will be kept in the project share. A reference and link to these datasets will be available in the GitHub repository.
- The code will be stored in GitHub privately until publication of results. Then, it will be published via the **GitHub - Zenodo integration, which will make the code citable**. A copy of the file with the model evaluation scores will also be provided in the Zenodo record.
- The size of the code will not exceed the Zenodo total size limit per record. Therefore, **no additional costs are foreseen**.

Research objective: develop model to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

Based on: https://doi.org/10.1016/j.jhydrol.2019.124351

Data sharing and reuse

Repositories, terms, contracts, licenses

Are any restrictions on data sharing required?

- No specific restrictions

How will you share data selected for sharing?

- Model and model evaluation resuls will be published in Zenodo, using the **GitHub - Zenodo integration**. Publication with Zenodo will ensure that the code and results are citable and accessible via its **persistent identifier** (DOI). **Metadata** will be provided at the time of publishing in Zenodo.
- The resulting Soil Surface Moisture will be published in **Pangaea**.
- A **CC-BY license** will be given to all outputs.

Research objective: develop model to estimate surface soil moisture (SSM) from satellite observations using a neural network model ("GRNN").

Based on: https://doi.org/10.1016/j.jhydrol.2019.124351

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PROJECT 3

- Study how ocean acidification interacts with light availability and how these impact not only calcification, but also other physiological responses.
  - Does elevated pCO2 affect calcification, coral biomass and tissue energy content, Symbiodinium density and chlorophyll concentration?
  - Are the effects of pCO2 on coral biomass and calcification modulated by light availability?

- Inspired on publication https://doi.org/10.1098/rsos.170683 and dataset https://doi.org/10.5061/dryad.5vg70

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Research objective: study of the interactive impacts of ocean acidification (OA) and light availability on coral tissue biomass composition and calcification.

Data summary / description

Purpose, origin, type, format & volume of the data

What data will you collect or create?

- The project will generate new experimental data. Ramets of the collected colonies of coral (*P. clavata*) will be fixed in tanks where they will be exposed to different conditions of light (low vs. high) and CO2 concentration (ambient vs. high), with an acclimatization period of 30 days. For each of the tanks, the environmental conditions of the tanks (independent variables) will be measured every three days during the experiment duration (32 days). Once the experiment is finished, the physiological response (dependent variables) under each of the conditions will be measured.
  - The different variables, format of the data and expected size is summarized in the table below.
  - Data from the experiments will be noted down in a laboratory notebook (paper) and regularly digitized into a corresponding spreadsheet (*.xlsx). This spreadsheet will be migrated to csv for the analysis. Csv is an open format, which guarantees its long term reusability.
- The data will be analysed in R to understand how the interaction of light conditions and concentration of CO2 impacts the physiological response of corals (*P. clavata*).

Dataset & type	Origin	Content/format	Estimated volume
Environmental conditions during acclimatization - experimental	Measured twice per week for 4 weeks in each of the tanks. These data is not part of the analysis, but will be monitored to discard interferences in the results. The values will be noted down on a laboratory notebook, and immediately digitized.	Paper format Digitized using excel and converted to csv	< 100 KB
Environmental conditions - experimental	Measured twice per week in each of the tanks from. Temperature, salinity, pH were obtained with sensors; alkalinity was measures with titration using a SOP. The values will be noted down on a laboratory notebook, and immediately digitized.	Paper format Measurement values will be digitized using an excel spreadsheet, which will be converted to csv (open) prior to analysis.	< 100 KB
Physiological response - experimental	Dependent variables measured at the end of the experiment (day 32) for all colonies in the different tanks. E.g. net calcification, photopigment densities, carbohydrates, proteins and lipids biomass, tissue energy content, *Symbiodinium* cell densities. These variables will be measured using appropriate SOPs, noted down on a laboratory notebook, and immediately digitized.	Paper format Measurement values will be digitized using an excel spreadsheet, which will be converted to csv (open) prior to analysis.	< 100 KB
Statistical analysis - development code	Multivariate response analysis will de done with R software (open). An R markdown file will also be produced to contextualize, document the analysis script and display the resulting figures.	R and R markdown (*.R, *.RMD)	<500 KB

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Research objective: study of the interactive impacts of ocean acidification (OA) and light availability on coral tissue biomass composition and calcification.

Documentation and metadata

Where and how, standards, ontologies, vocabularies

How will you document the data?

Documentation

- **Data collection**: a separate document will be prepared indicating both the measurement equipment and laboratory Standard Operating Procedures (SOP) used during the experimental phase of the project. Where possible, references or links to the technical specification of the equipment and the SOPs will be provided.
- The information of the **laboratory notebooks** will be digitized. Besides parameter values, other relevant information that was observed for a particular day & tank will be noted down.
- A **codebook** will be prepared, indicating the what each variable means, units and possible values when applicable. This will be done for the following:
  - File with data environmental variables
  - File with data about physiological reponse
  - Output file of the multivariate response analysis.
- An **R markdown** document will also be created. This document will contain contextual information about the project (e.g. experimental design), and integrate snippets of codes, resulting figures and detail explanations of the analysis and results.

Metadata

- We plan to make the data and accompanying documentation available in Pangaea or, preferably, another discipline-specific repository.
  - Pangaea uses the ISO 19115 metadata standard for geospatial data and provides the possibility to describe the data according to characteristics that increase their findability (e.g. methods, spatial and temporal coverage, topic). Besides, each dataset contains all the contextual information necessary, such as author, abstract, related publications or links, etc.
  - In addition, data files in Pangaea are assigned a metaheader with the fields: citation, reference, project(s), coverage, parameters, license, number of data points. (See: https://wiki.pangaea.de/wiki/Metaheader).

Research objective: study of the interactive impacts of ocean acidification (OA) and light availability on coral tissue biomass composition and calcification.

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Data storage during the project

Security, back-up, transfer & access

How will you store and backup data during research?

- During the project, data obtained from the experiments and digitized from the lab notebook will be kept in excel spreadsheets on the project Sharepoint.
- Sharepoint is part of the storage infrastructure provided by the University ICT department, which offers technical support. The size of the data files is small and no additional resources will be needed.
- Code will be developed locally and regularly pushed to a dedicated GitHub repository, enabling version control of the analysis scripts.
- **Data organization**. Files in the project site will be organized according to the following hierarchy:
    - 01_Literature
    - 02_Experimental_phase
        - Env_conditions.xlsx
        - Physiol_response.xlsx
        - SOPs
        - Instruments_SOPs.docx
    - 03_Analysis
        - Input
            - Env_conditions.csv
            - Physiol_response.csv
        - Temp: folder to store temporal or intermediate outputs while developing analysis scripts. All files in this folder will be preceded by the date in YYYYMMDD format, so they can be linked to specific versions of the code.
        - Output
            - Tables: the content of the csv files with output tables will be indicated by making use of the same variable codes as in the analysis script.
            - Figures: name of the figures will be based in the x and y variable codes used in the analysis script.
    - 04_Scripts
        - Will be managed in GitHub, but a final version will be downloaded and kept together with the rest of the project files.
    - 05_Reporting
        - Article
            - Draft
            - Submission
        - Presentations
            - File names will follow: YYYYMMDD_EventName_PresentationTitle
        - Conferences

Research objective: study of the interactive impacts of ocean acidification (OA) and light availability on coral tissue biomass composition and calcification.

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UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY

rdm.support@ugent.be

@UGentRDM

ugent.be/en/research/datamanagement

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EXAMPLE DMPS

- UGent examples: https://www.ugent.be/en/research/datamanagement/before-research/datamanagementplan.htm#collapse5

- Liber DMP catalogue: https://libereurope.eu/dmpcatalogue/
- Horizon 2020 examples: https://phaidra.univie.ac.at/search#?page=1&pagesize=10&collection=o:1140797

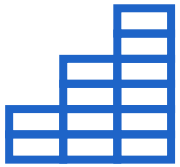
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WHICH NEEDS TO CONSIDER?



COLLABORATION

- → Who is allowed to access the data?
- → How to share data?



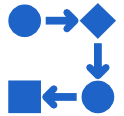
DATA VOLUME

- → Which storage solution offers sufficient capacity?
- → What to do when I have large volumes of data (> 2Tb)?



CONFIDENTIALITY

- → Which data security measures to apply?
- → Can I re-use 3rd party data?
- → Which privacy regulations to comply with?
- → How can I allow for IP rights protection?



ACTIVE

- → How frequently do I need to access the data?
- → Who should be able to access and use the data?
- → How to understand the data?
- → Can I reproduce data from some processing stages?



DOMAIN

- → Which standard data formats are used for domain-specific data types?
- → Which documentation methods are common in my scientific domain?

PROCESSING & ANALYSIS SCRIPTS

- Experimental design
- Seawater chemistry
- Physiological Responses
- Assumptions of ANOVA:
- Statistical Models
- Principal Component Analysis
- Figures: PCA
- Univariate Models: lmer

Elevated pCO_2 affects tissue biomass composition, but not calcification, in a reef coral under two light regimes

Code

CB Wall, Hawai'i Institute of Marine Biology. Wall et al. (2017) Royal Society Open Science
9/29/2017

Experimental design

- Treatments were created in 24 tanks of 6 replicate treatents:
- Low Light-Ambient pCO_2 (LL-ACO_2), Low light-High pCO_2 (LL-HCO_2), High light-Ambient pCO_2 (HL-ACO_2), and High light-High pCO_2 (HL-HCO_2).
- two tanks malfunctioned and the the final tank replicates were 6 tanks for all treatments, but 4 replicate tanks for HL-HC treatment
- Experimental factors (pCO_2 and light) were treated as fixed effects. Tank replicates were treated as random effects nested within treatment. Coral colony was treated as a random effect.

Seawater chemistry

Exploring seawater chemistry data, generating data summaries to be used in tables and figures. Statistical tests performed to test for differences in pH, total alkalinity, and pCO_2 among all treatment tanks and among replicate treatment tanks within a CO_2 treatment (ambient or high).

Attach packages

Code

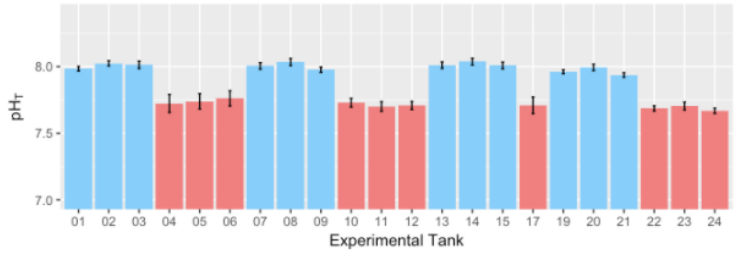
Attach the first dataset for seawater chemistry.

Code

pH and CO_2 treatment figure

Generate a figure of mean +/- SE pHT and pCO_2 for each treatment tank.

Code



Hide

```
# set working directory to 'figures' for figure export

setwd('markdown/figures')
#######################################
## EXPERIMENT:  Figures of CO2 and pHT ##
#######################################
colors=c("lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightcoral", "lightcoral","lightskyblue", "lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightcoral", "lightcoral","lightskyblue", "lightskyblue",
"lightskyblue", "lightcoral","lightskyblue", "lightskyblue", "lightskyblue", "lightcoral","lightcoral", "lightcoral")

# plot for all tank pH and pCO2

################
###### pH ######
################
mean<-tapply(SWdata$pHT, SWdata$Tank, mean); se<-tapply(SWdata$pHT, SWdata$Tank, std.error); pH<-data.frame(mean, se); pH["Tank"] <- c("01", "02", "03", "04", "05","06","07","08","09","10","11","12","13","14","15","17","19","20","21","22","23","24"); colnames(pH) <- c("mean", "se","Tank")

Fig.pH <- ggplot(pH, aes(x=Tank, y=mean)) +
  geom_bar(position="dodge", stat="identity", fill=colors) +
  geom_errorbar(aes(ymin=pH$mean - pH$se, ymax=pH$mean + pH$se), width=0.1) +
  xlab(expression("Experimental Tank")) +
  ylab(expression(pH[T])) +
  scale_y_continuous(limits=c(7.0,8.4),oob = rescale_none)

Fig.pH # gives summary figure of tank pH mean +/-SE across all tanks
```

WHERE?

(Online) code notebooks (e.g. Jupyter notebook, R markdown)

- Create and share documents that contain live code, equations, visualizations and explanatory text
- Open, reproducible research

https://doi.org/10.5061/dryad.5vg70

More on reproducibility: https://the-turing-way.netlify.app/reproducible-research/reproducible-research.html

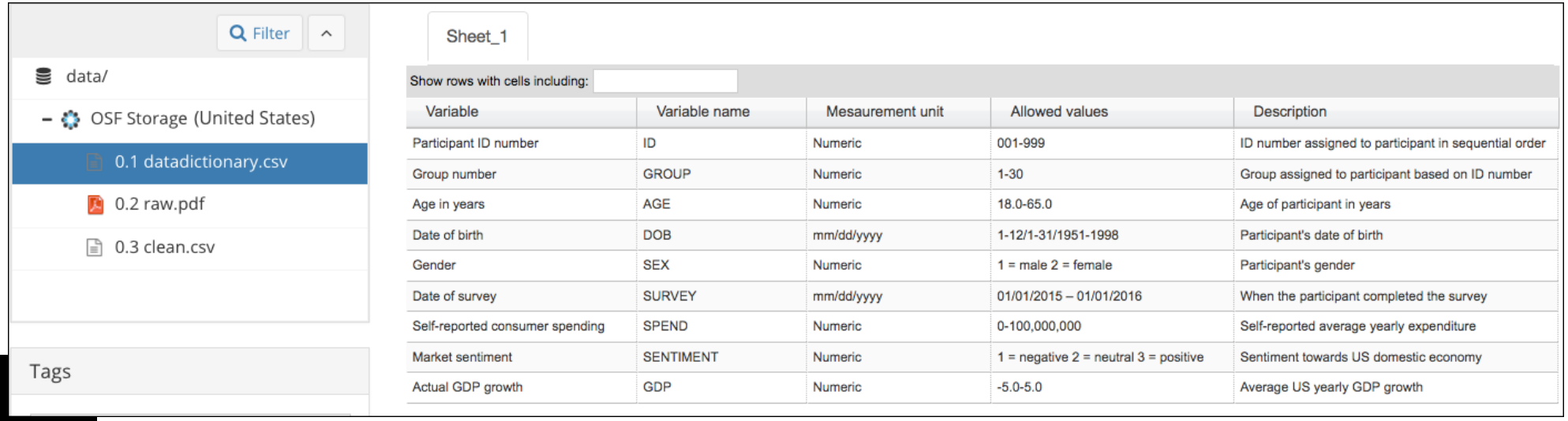
VARIABLE OR ITEM LEVEL DOCUMENTATION

WHAT?

- Code books or variable dictionaries
  - Variable labels & meaning
  - Attributes, expected values, units
  - Codes, abbreviations, classification schemes
  - Variable derivation & processing steps: what & why

WHERE?

- Embedded in file: content limitation and conversion risks
- Separate file



Variable	Variable name	Mesaurement unit	Allowed values	Description
Participant ID number	ID	Numeric	001-999	ID number assigned to participant in sequential order
Group number	GROUP	Numeric	1-30	Group assigned to participant based on ID number
Age in years	AGE	Numeric	18.0-65.0	Age of participant in years
Date of birth	DOB	mm/dd/yyyy	1-12/1-31/1951-1998	Participant's date of birth
Gender	SEX	Numeric	1 = male 2 = female	Participant's gender
Date of survey	SURVEY	mm/dd/yyyy	01/01/2015 – 01/01/2016	When the participant completed the survey
Self-reported consumer spending	SPEND	Numeric	0-100,000,000	Self-reported average yearly expenditure
Market sentiment	SENTIMENT	Numeric	1 = negative 2 = neutral 3 = positive	Sentiment towards US domestic economy
Actual GDP growth	GDP	Numeric	-5.0-5.0	Average US yearly GDP growth

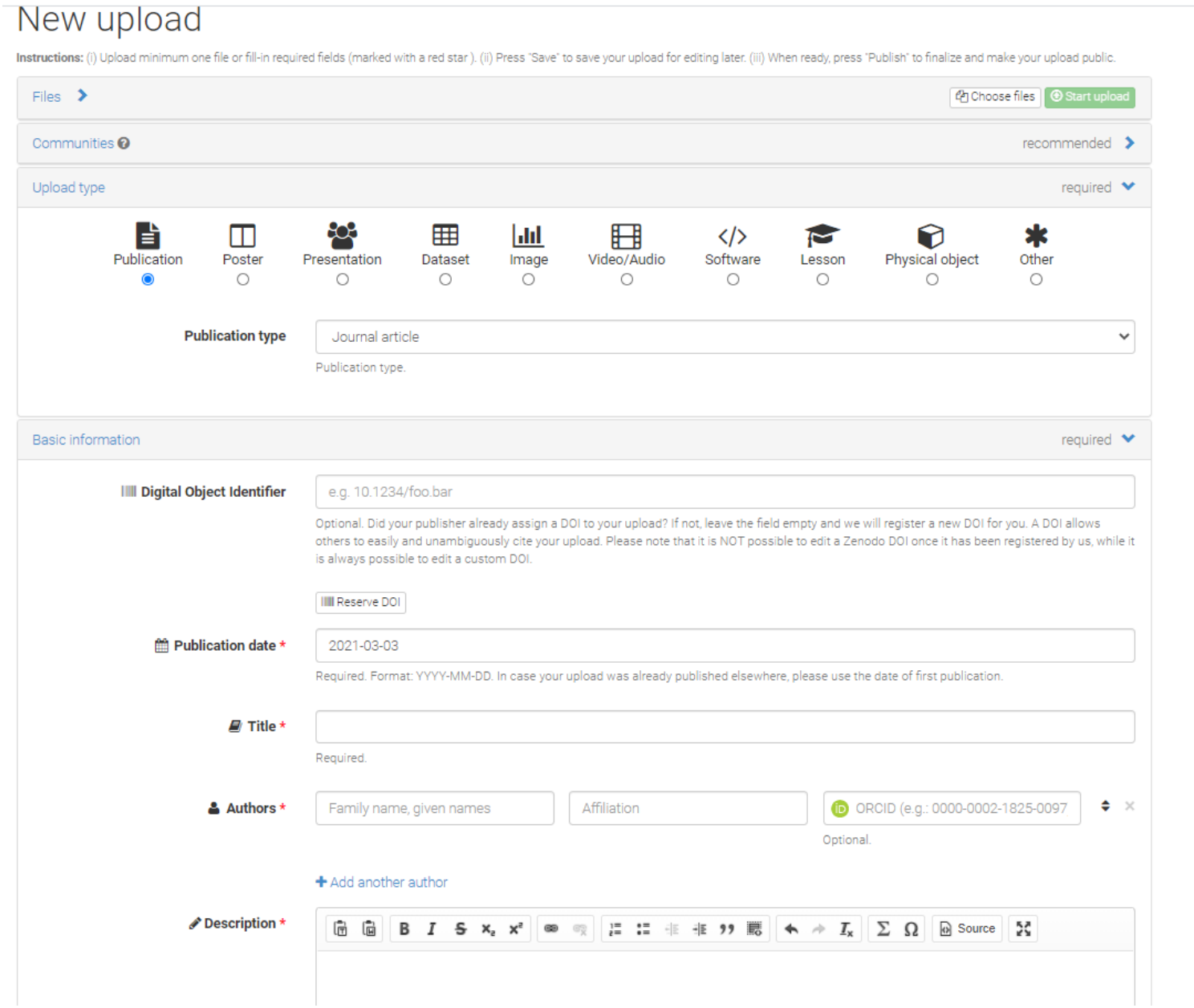
How to Make a Data Dictionary

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METADATA IN DATA REPOSITORIES

- Repositories: infrastructure to allow us to share & find data
- Provide **services to create & manage metadata**
- Typically, by filling a form with
  - Pre-defined fields
  - Lists of controlled terms to choose from (e.g. standard licenses)
- Metadata adheres to a specific metadata standard (interoperability)

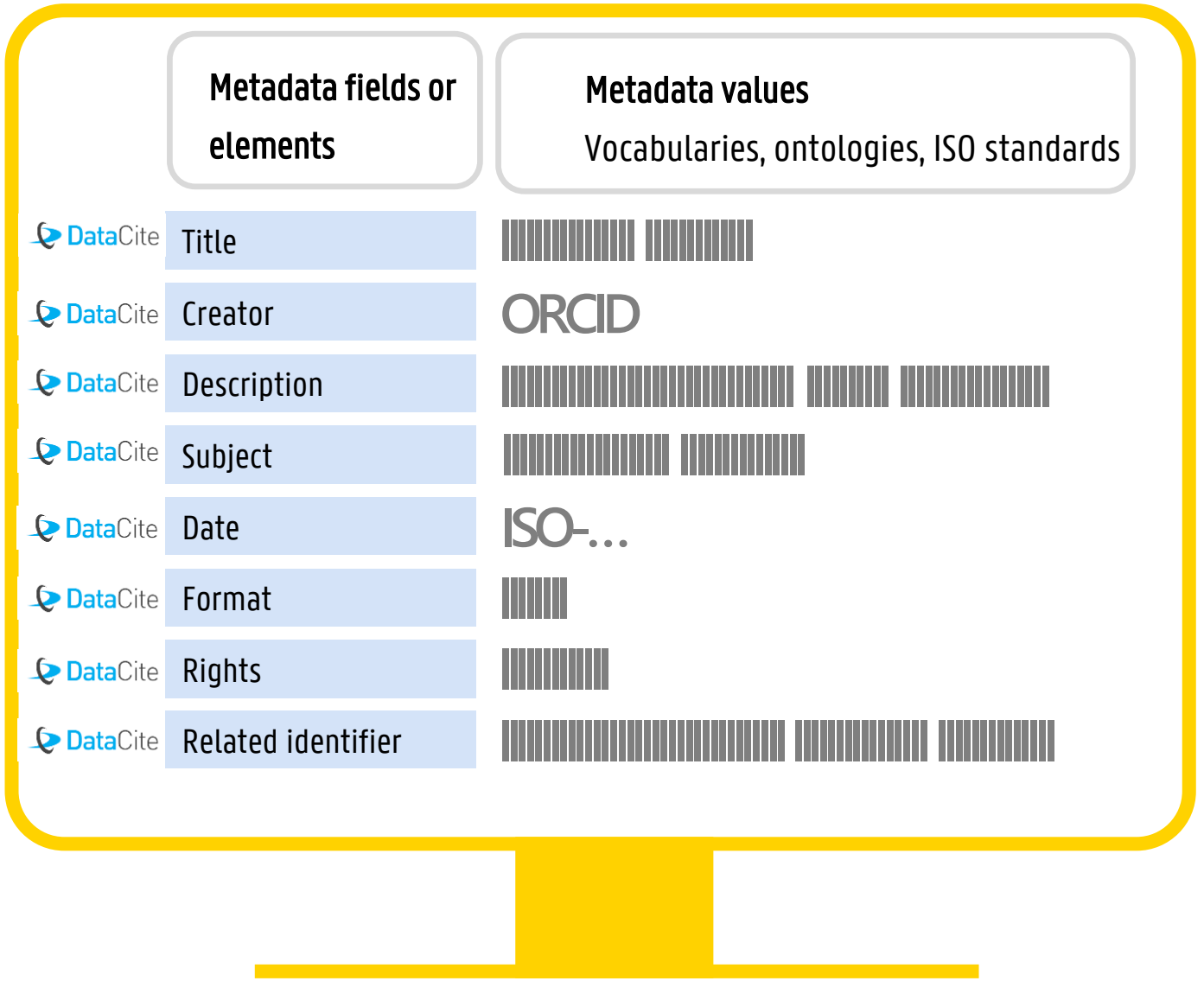
Zenodo data upload form



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VOCABULARIES AND STANDARDS

STANDARDIZED METADATA



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HOW TO SELECT A DATA REPOSITORY?

Recommended repositories

RESEARCH DOMAIN'S CONSENSUS



RESEARCHERS' BEST PRACTICES



JOURNAL RECOMMENDATIONS



data repositories
recommended by Nature

DATA REPOSITORY REGISTRIES

re3data

FAIRsharing

re3data.org demo

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HOW TO SELECT A DATA REPOSITORY?

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

seadatanet | Search

http://www.re3data.org

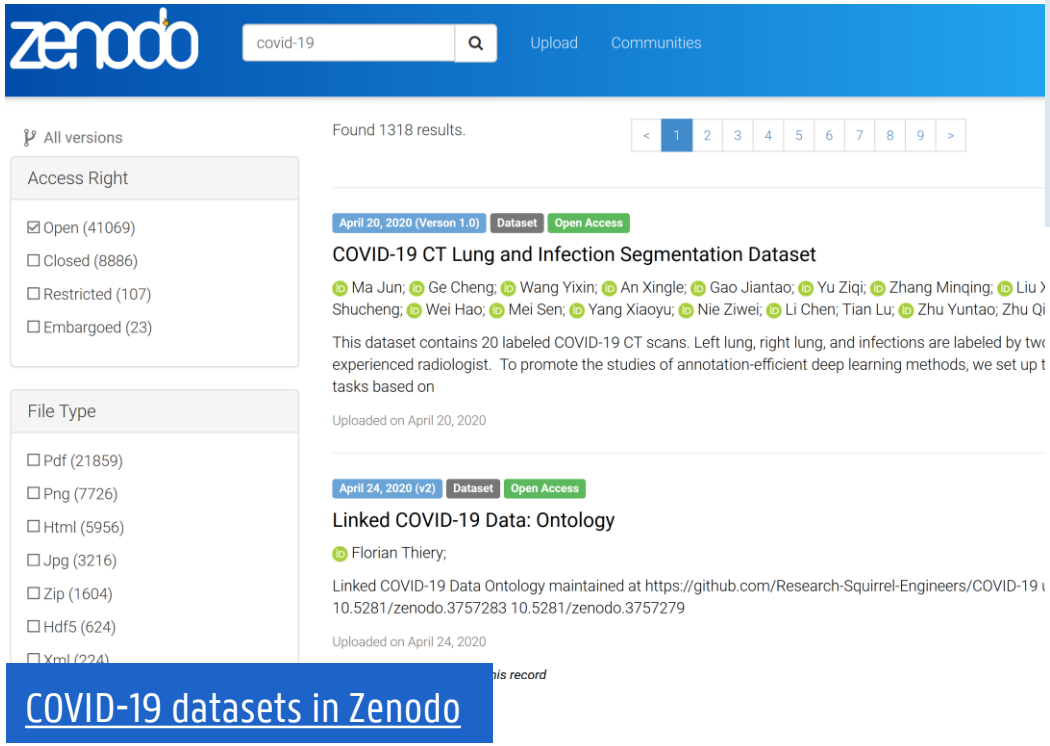
zenodo
B2SHARE
EUDAT
DRYAD
BioStudies.
Global Biodiversity Information Facility
Free and Open Access to Biodiversity Data
ENA
European Nucleotide Archive
DataONE
Data Observation Network for Earth
Edinburgh DataShare
DataSTORRE
DataSTORRE: Stirling Online Repository for Research Data.

eneral purpose | Domain-specific | Institutional

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Watch a Re3data demo: https://www.fosteropenscience.eu/content/re3data-demo

GENERIC VS. DOMAIN-SPECIFIC REPOSITORIES



COVID-19 datasets in Zenodo



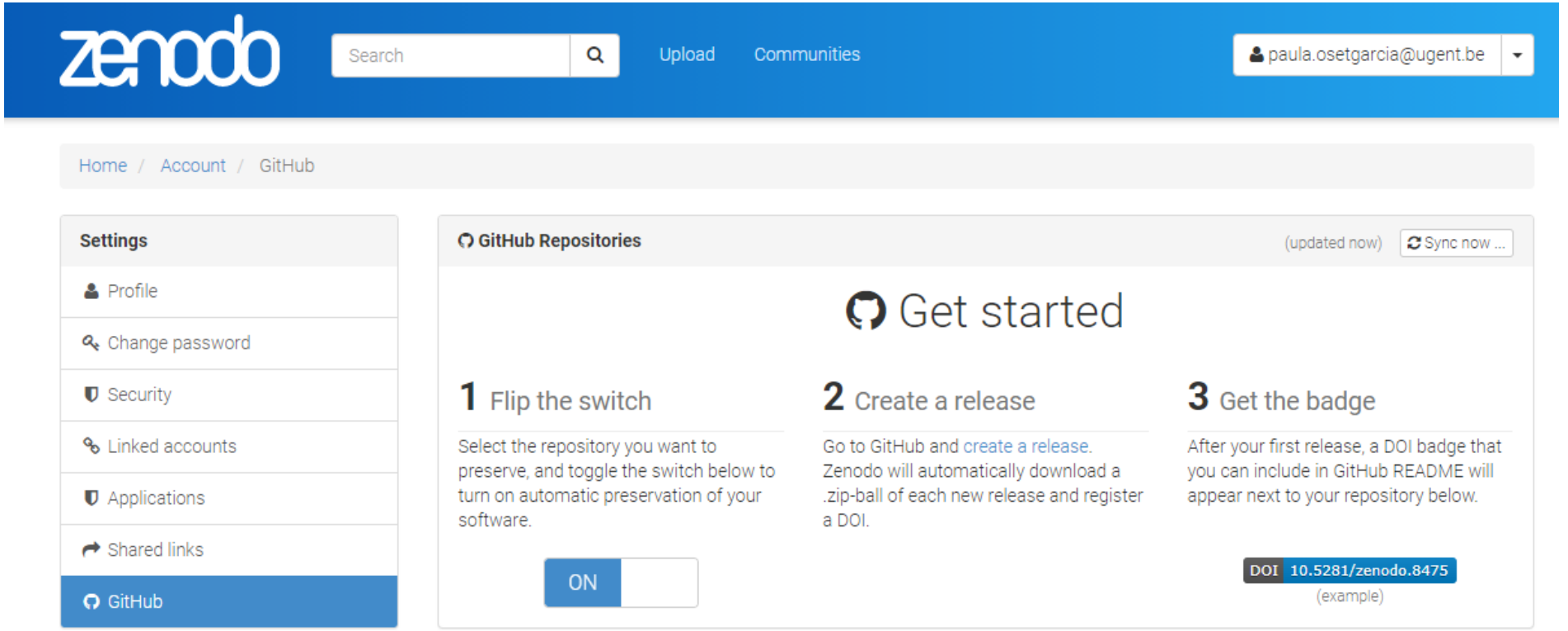


COVID-19 data portal

SHARING CODE AND MAKING CODE CITABLE

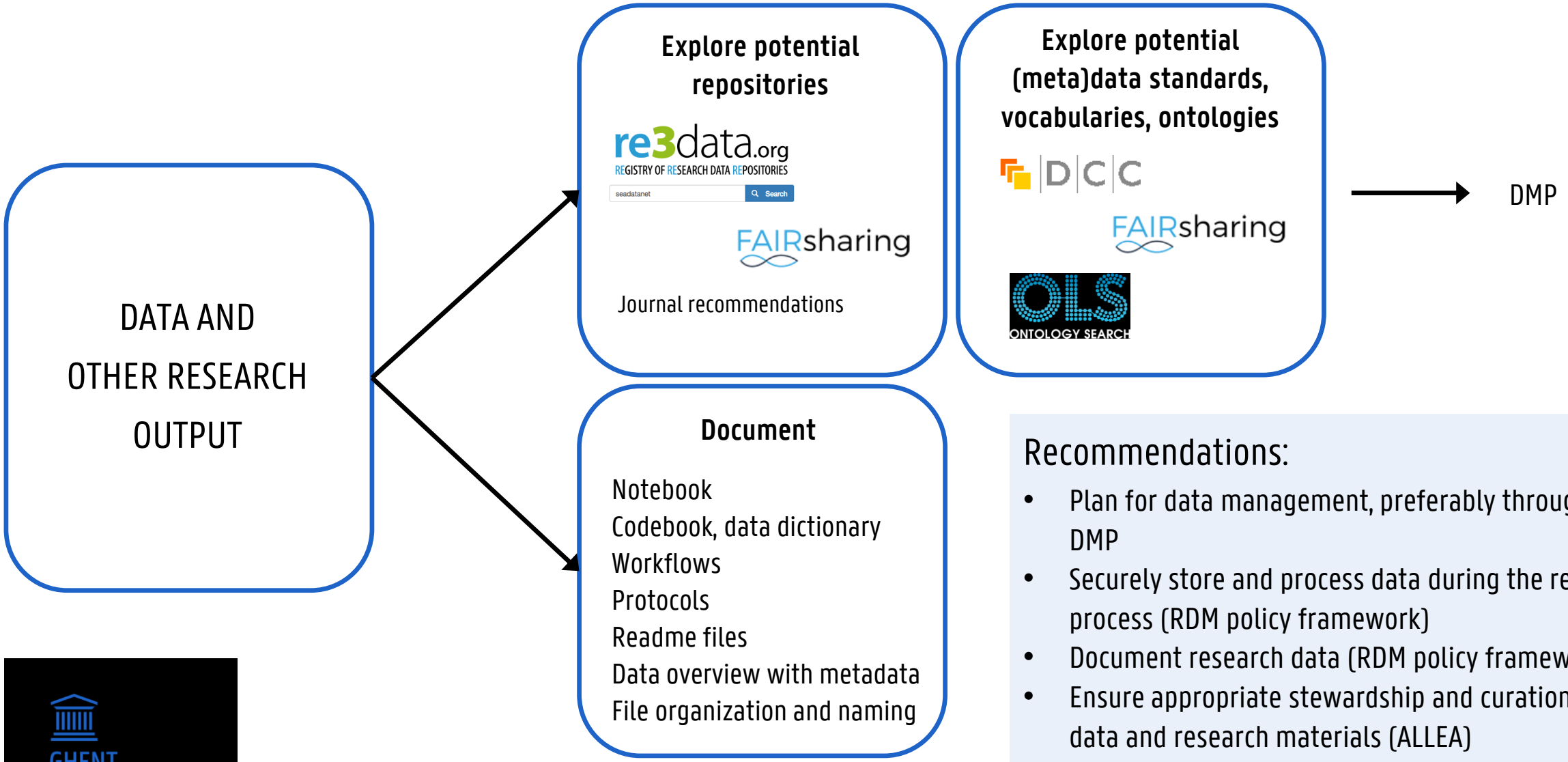
GitHub – Zenodo integration: https://guides.github.com/activities/citable-code/





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MAXIMISING DATA RE-USE DURING RESEARCH



Recommendations:

- Plan for data management, preferably through a DMP
- Securely store and process data during the research process (RDM policy framework)
- Document research data (RDM policy framework)
- Ensure appropriate stewardship and curation of all data and research materials (ALLEA)

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MAXIMISING DATA RE-USE AFTER RESEARCH

DATA AND
OTHER RESEARCH
OUTPUT
+
DOCUMENTATION
METADATA

Option 1: Domain-specific repository (preferred)



Global Biodiversity Information Facility
Free and Open Access to Biodiversity Data

ads ARCHAEOLOGY DATA SERVICE

DataONE
Data Observation Network for Earth

Option 2: Generic repository

zenodo

figshare

DRYAD

OSF

Institutional repository

COMING SOON

Register **metadata** in Biblio with PID if available

Recommendations:

- Share within UGent for scientific review and more widely if possible (RDM policy framework)
- Share as openly as possible, as closed as necessary, in line with the FAIR principles (ALLEA)
- Use domain-specific repository for sharing outside of UGent when available

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